

Eye-specific active zone clustering underlies synaptic competition in the developing visual system

Chenghang Zhang, Tarlan Vatan, Colenso M Speer 

Department of Biology, University of Maryland, College Park, United States

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

Reviewed Preprint

v3 • September 8, 2025

Revised by authors

Reviewed Preprint

v2 • December 16, 2024

Reviewed Preprint

v1 • October 23, 2023

eLife Assessment

This is a **useful** analysis of STORM data that characterizes the clustering of active zones in retinogeniculate terminals across ages and in the absence of retinal waves. The design makes it possible to relate fixed time point structural data to a known outcome of activity-dependent remodeling. However, the evidence is **incomplete**, weakening the claims the authors make regarding how activity influences the clustering of these synapses. This basic criticism has not improved with revisions.

<https://doi.org/10.7554/eLife.91431.3.sa4>

Abstract

Spatially clustered synaptic inputs enable local dendritic computations important for learning, memory, and sensory processing. In the mammalian visual system, individual retinal ganglion cell (RGC) axons form clustered terminal boutons containing multiple active zones onto relay cell dendrites in the dorsal lateral geniculate nucleus (dLGN). This mature architecture arises through the addition of release sites, which strengthens selected afferents while weaker inputs are pruned. Following eye-opening, spontaneous activity and visual experience promote synaptic refinement and bouton clustering after binocular inputs have segregated. However, anatomical changes in release site addition and spatial patterning during earlier stages of eye-specific competition are not well understood. To investigate this, we examined the spatial organization of eye-specific active zones in wild type mice and a mutant line with disrupted cholinergic retinal waves. Using volumetric super-resolution single-molecule localization microscopy and electron microscopy, we found that individual retinogeniculate boutons begin forming multiple nearby presynaptic active zones during the first postnatal week. Both eyes generate these “multi-active-zone” (mAZ) inputs throughout refinement, but the dominant-eye forms more numerous mAZ contacts, each with more active zones and larger vesicle pools. At the height of competition, the non-dominant-eye projection adds many single active zone (sAZ) synapses. Mutants with abnormal cholinergic retinal waves still form mAZ inputs, but develop fewer sAZ synapses and show reduced synapse clustering in projections from both eyes. Together, these findings reveal activity-dependent, eye-specific differences in release site addition during synaptic competition in circuits essential for visual perception and behavior.

Introduction

A key mechanism for neuronal signal processing is the formation of spatially clustered synapses that facilitate local computations within individual dendrites ^{1–3}. Through biophysical signal integration mechanisms, neighboring synaptic inputs play critical computational roles in learning, memory, and sensory processing underlying cognition and behavior ^{4–7}. During circuit development, both spontaneous and sensory-driven neural activity regulate synaptic clustering, stabilizing some synapses and eliminating others to establish mature connectivity patterns ^{6–8}.

The refinement of retinal inputs to the dorsal lateral geniculate nucleus (dLGN) of the thalamus is a model example of activity-dependent synaptic clustering ⁹. Over development, boutons within the terminal arbors of individual retinogeniculate axons cluster progressively ^{10–13}, creating multiple synaptic contacts onto individual postsynaptic targets ^{14–16}. Concurrently, developing RGC boutons add presynaptic release sites such that some mature retinogeniculate terminals contain several dozen active zones (AZs) ^{16, 17}. The formation of clustered boutons with multiple release sites subserves a critical “driver” function of retinogeniculate input ^{18, 19}.

Retinogeniculate bouton clustering depends on visual experience, and dark-rearing after eye-opening reduces clustering within individual axon arbors ¹². However, less is known about synaptic spatial relationships and activity-dependent release site refinement during eye-specific competition before eye-opening. During early axon ingrowth, individual RGCs extend sparse side branches in the incorrect eye-specific territory ^{20–22}. These transient branches form synapses that contain fewer presynaptic vesicles than those made by the same axon in the correct eye-specific domain ²³.

Functional recordings after eye-opening show that while many RGC inputs are pruned, the remaining inputs strengthen by adding release sites ²⁴. This leads to glutamate spillover and cross talk between RGC inputs that increases postsynaptic relay neuron excitability ²⁵. Thus, early eye-specific differences in release site number or spacing may contribute to binocular input refinement prior to eye-opening.

In a previous study from our laboratory, we used volumetric stochastic optical reconstruction microscopy (STORM), anterograde tract tracing, and immunohistochemical labeling of synaptic proteins to show that dominant-eye synapses contain more total vesicles and have more vesicles near the AZ (putative docking) compared with non-dominant-eye synapses ²⁶. However, we did not examine whether eye-specific inputs differ in AZ number or AZ spatial proximity (clustering) during synaptic competition. To address this gap, we reanalyzed our published dataset, focusing on eye-specific AZ spatial relationships. We found that projections from each eye form a subset of retinogeniculate inputs in which several (2–4) AZs share a common cluster of presynaptic vesicles. We refer to these synapses as multi-active-zone (mAZ) inputs. All other retinogeniculate inputs contained one single active zone and its associated vesicle cluster, and we term these single-active-zone (sAZ) synapses.

During eye-specific synaptic competition, the dominant-eye projection formed more mAZ inputs, each with more AZs and a larger presynaptic vesicle pool compared to the non-dominant eye projection. Similarly, the dominant-eye had higher vesicle signal at sAZ inputs. At the peak of synaptic competition midway through the first postnatal week, the non-dominant-eye formed numerous sAZ inputs, equalizing the global synapse density between the two eyes. These eye-specific AZ patterns were disrupted in a mutant mouse line with abnormal stage II cholinergic

retinal waves and retinogeniculate segregation defects. Our results demonstrate that eye-specific presynaptic release site clustering arises before eye-opening and is impacted by defects in spontaneous retinal activity.

Results

Retinogeniculate inputs form multiple active zones during eye-specific competition

To investigate active zone refinement during eye-specific segregation, we reanalyzed a volumetric super-resolution imaging dataset previously published by our laboratory²⁶. We used volumetric STochastic Optical Reconstruction Microscopy (STORM)²⁷ to image the dLGNs of wild-type (WT) mice at three postnatal ages (P2, P4, and P8) (**Fig. 1A**). We labeled eye-specific inputs by monocular injection of Alexa Fluor-conjugated cholera toxin subunit B tracer (CTB) together with immunostaining for presynaptic Bassoon, postsynaptic Homer1, and presynaptic vesicular glutamate transporter 2 (VGluT2) proteins (**Fig. 1B**). We collected separate image volumes (~45K μm^3 each) from three biological replicates at each age. To assess the impact of spontaneous retinal activity on synaptic development across the same time period, we performed identical experiments in $\beta 2$ -knockout ($\beta 2\text{KO}$) mice lacking the beta 2 subunit of the nicotinic acetylcholine receptor, a mutation that disrupts spontaneous cholinergic retinal wave activity, eye-specific segregation, and retinogeniculate synapse development^{22, 26, 28–36} (**Fig. 1A**). Because eye-specific segregation is incomplete until ~P8, we limited our re-analysis to the future contralateral eye-specific region of the dLGN, which is reliably identified across all stages of postnatal development (**Fig. 1A**, see also Materials and Methods).

By analyzing synapses in the contralateral dLGN from eighteen mice across three ages and two genotypes (Table S1), STORM revealed two classes of retinogeniculate inputs distinguished by active zone (AZ) number (**Fig. 1B**). We defined each retinogeniculate input as a single contiguous VGluT2 cluster together with all its associated presynaptic (Bassoon) and postsynaptic (Homer1) paired synaptic labels. Using this definition, inputs that had multiple (2–4) Bassoon AZs were classified as multi-active-zone (mAZ) inputs, while those with a single Bassoon AZ were designated single-active-zone (sAZ) inputs (**Fig. 1B**). Most mAZ inputs contained two AZs (~70–90%, varying with age, genotype, and eye-of-origin); smaller proportions contained three AZs (~10–20%) or four or more AZs (<5%) (Table S1).

Each mAZ input could be a single terminal bouton with several AZs, or a cluster of sAZ synapses within separate boutons^{10, 11, 15, 16, 37}. To address this, we used electron microscopy (EM) to image retinogeniculate terminals in the dLGN at P8. We generated a transgenic line expressing mitochondrial matrix-targeted dimeric dAPEX2 reporter³⁸ in ipsilaterally-projecting RGCs^{39–41}, providing unambiguous mitochondrial labeling in ipsiRGC axons. EM images confirmed the presence of individual retinogeniculate boutons with multiple active zones, consistent with our STORM data (**Fig. 1C**). Previous EM reconstructions of retinogeniculate inputs reported no evidence of RGC bouton convergence at the end of the first postnatal week¹¹. Together, these results suggest that mAZ inputs in STORM images are single RGC terminals that house several closely spaced release sites. Hereafter, we use the word “synapse” only to refer to each partnered active zone (Bassoon/Homer1 pairs); mAZ inputs contain several synapses, while each sAZ input is one synapse.

Changes in eye-specific input density during synaptic competition

In our previous analysis, we reported global eye-specific synapse densities that reflected the combined mAZ and sAZ inputs. Dominant-eye synapse density was greater than that of the non-dominant eye at P2 and P8, but eye-specific synapse densities were equivalent at P4 during the

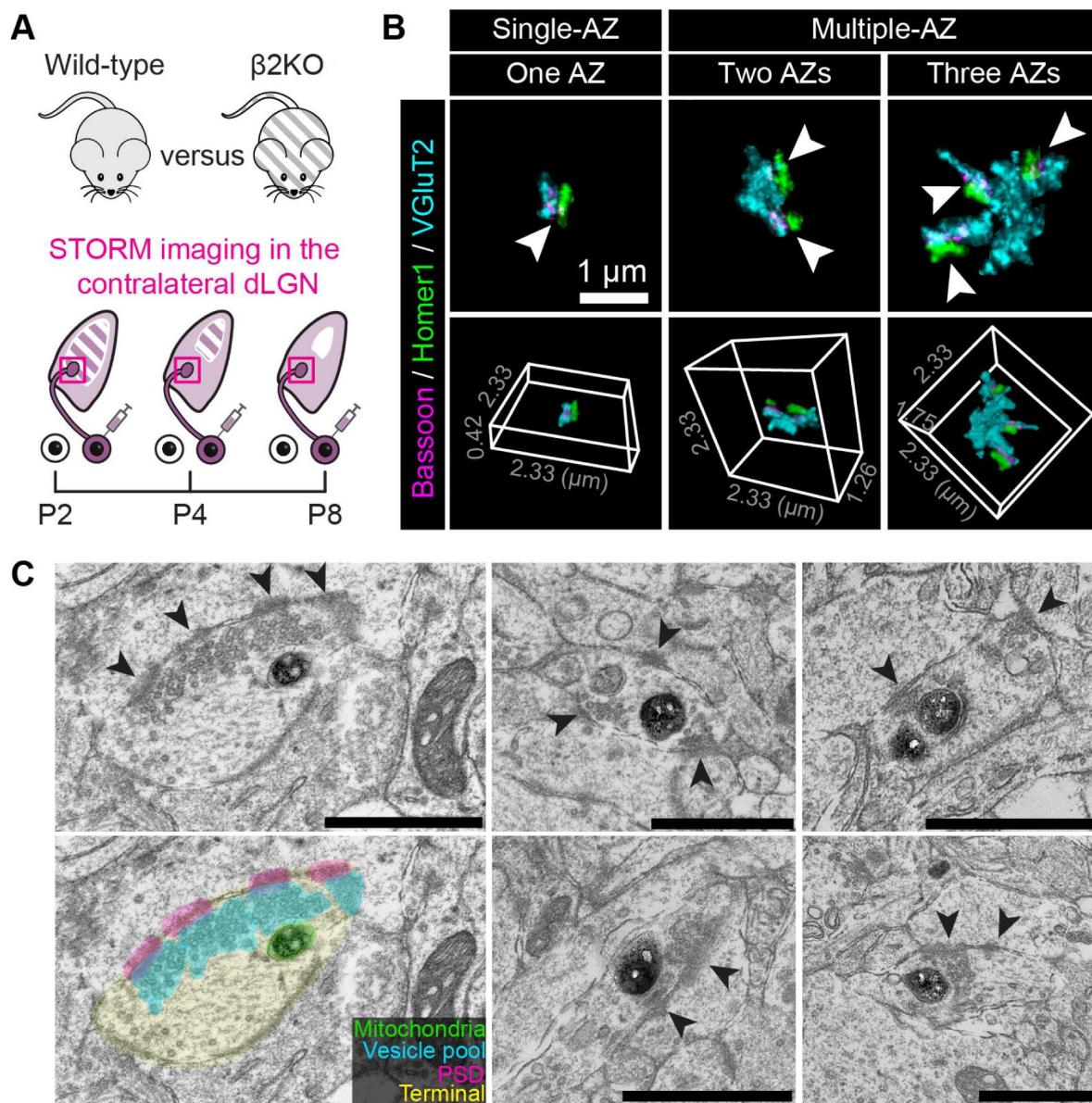


Fig. 1

Retinogeniculate boutons form multiple active zones during eye-specific competition.

(A) Experimental design. CTB-Alexa 488 was injected into the right eye of wild-type and $\beta 2\text{KO}$ mice. One day after the treatment, tissue was collected from the left dLGN at P2, P4, and P8. Red squares indicate the STORM imaging regions that were analyzed. (B) Representative examples of individual sAZ and mAZ inputs, with corresponding active zone counts ranging from one to three. Upper panels show Z-projections of inputs and lower panels show the corresponding 3D volume. Arrowheads point to individual Bassoon clusters (active zones) paired with postsynaptic Homer1 labels within each input. All examples are from a WT P8 sample. (C) Electron micrographs of mAZ retinogeniculate inputs in a P8 $\text{ET33}^{\text{Cre}}::\text{ROSA26}^{\text{LSL-Matrix-dAPEX2}}$ mouse. Darkly-stained dAPEX2(+) mitochondria are present within ipsilaterally-projecting RGC terminals. Arrowheads point to electron-dense material at the postsynaptic density, apposed to individual active zones with clustered presynaptic synaptic vesicles.

peak of synaptic competition²⁶. To determine how these two input classes evolve during competition, we quantified the densities and percentages of mAZ and sAZ inputs in WT and β 2KO mice (**Fig. 2A**). Eye-of-origin for each retinogeniculate input was assigned by colocalizing CTB signal with VGlut2 (**Fig. 2B**). Binocular CTB control injections showed that anterograde tracing labeled >97% of retinogeniculate synapses at P4 and P8, ensuring accurate eye-specific assignment²⁶. Within the contralateral eye-specific region of the dLGN, CTB(+) VGlut2 clusters were classified as “dominant-eye” inputs and CTB(-) VGlut2 clusters as “non-dominant eye” inputs (**Fig. 2B**).

After assigning eye-of-origin to each retinogeniculate synapse, we found that the density of CTB(+) mAZ inputs was significantly higher than CTB(-) mAZ inputs at every age in WT mice (**Fig. 2C**, top left; Fig. S1, top). β 2KO mice also developed eye-specific differences in mAZ input density (**Fig. 2C**, bottom left; Fig. S1, bottom). For WT sAZ synapses, the dominant-eye had a higher synapse density at P2 and P8. At P4, however, sAZ synapse density was equivalent between the eyes (**Fig. 2C**, top right).

This pattern resulted from non-dominant eye sAZ synapse addition, which equalized the global input density between the eyes at P4 as we reported previously (5/95% confidence interval, -0.014-0.011 synapses / μm^3). In β 2KO mice, the non-dominant eye formed fewer sAZ synapses, trending toward an eye-specific difference. (**Fig. 2C**, bottom right; $p < 0.05$, power=0.62; see Tables S2/S3). Thus, β 2KO mice with disrupted retinal activity maintain a higher mAZ input fraction in the dominant-eye projection, but sAZ synapse addition is reduced at the peak of competition.

Multi and single active zone inputs from the dominant eye show increased vesicle pool size and vesicle proximity to the active zone

We previously reported a dominant-eye bias in VGlut2 volume when considering all retinogeniculate inputs²⁶. Here, we assessed presynaptic maturation separately in sAZ and mAZ inputs by measuring their total VGlut2 volume. In WT mice, both mAZ (**Fig. 3A**, left) and sAZ (**Fig. 3B**, left) inputs showed significant eye-specific volume differences at each age. In the middle of eye-specific competition at P4 in WT mice, the median VGlut2 cluster volume in dominant-eye mAZ inputs was 3.72 fold larger than that of non-dominant-eye inputs (**Fig. 3A**, left). In contrast, β 2KO mice showed a smaller 1.1 fold difference at the same age (**Fig. 3A**, right panel). For sAZ synapses at P4, the magnitudes of eye-specific differences in VGlut2 volume were smaller: 1.35-fold in WT (**Fig. 3B**, left) and 0.41-fold in β 2KO mice (**Fig. 3B**, right). Thus, both mAZ and sAZ input size favors the dominant eye, with larger eye-specific differences seen in WT mice (see Table S3).

In addition to total vesicle pool volume, we quantified VGlut2 volume within a 70 nm shell around each AZ in mAZ and sAZ inputs, considering this a proxy for docked vesicles (Fig. S2A-C)²⁶. In WT mice at P4, dominant-eye inputs showed greater per-AZ vesicle volume than non-dominant inputs in both mAZ and sAZ categories (Fig. S2B, left). These eye-specific differences were absent in β 2KO mice (Fig. S2B, right).

Moreover, at all ages and in both genotypes, vesicle volume within the 70 nm shell did not differ between mAZ and sAZ inputs (Fig. S2A-C). This shows that vesicle docking at release sites favors the dominant-eye as we previously reported²⁶, but is similar for like-eye type inputs regardless of AZ number.

Vesicle pool size scales with active zone number

Because mAZ inputs showed greater total VGlut2 volume than sAZ synapses, yet exhibited comparable vesicle docking, the disparity could reflect a scaling effect of vesicle pool size with increased AZ number. To evaluate how presynaptic vesicle pool volume scales with AZ number,

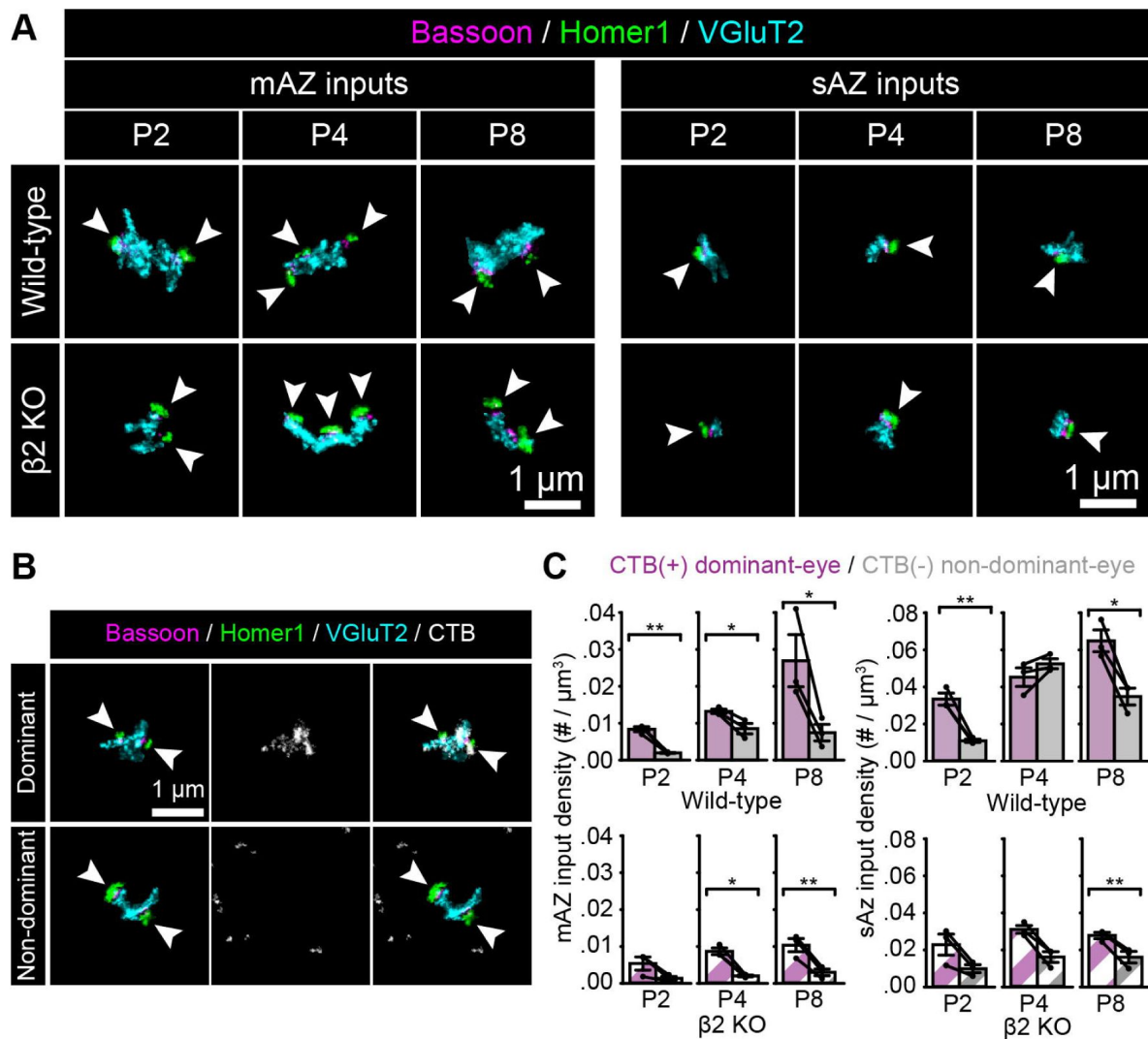


Fig. 2

Changes in eye-specific input density during synaptic competition.

(A) Representative Z-projection images of mAZ and sAZ inputs across ages and genotypes. Arrowheads point to individual Bassoon/Homer1 cluster pairs indicating release sites. (B) Representative CTB(+) dominant-eye (top panels) and CTB(-) non-dominant-eye (bottom panels) mAZ inputs in a WT P8 sample, showing synaptic (left panels), CTB (middle panels), and merged labels (right panels). Arrowheads point to individual Bassoon/Homer1 paired clusters. (C) Eye-specific mAZ (left) and sAZ (right) input density across development in WT (top panels) and $\beta 2$ KO mice (bottom panels). Black dots represent mean values from separate biological replicates and black lines connect eye-specific measurements within each replicate (N=3 for each age and genotype). Error bars represent group means $\pm$ SEMs. Statistical significance between eye-specific measurements was assessed using two-tailed paired T-tests at each age. * $p < 0.05$, ** $p < 0.01$.

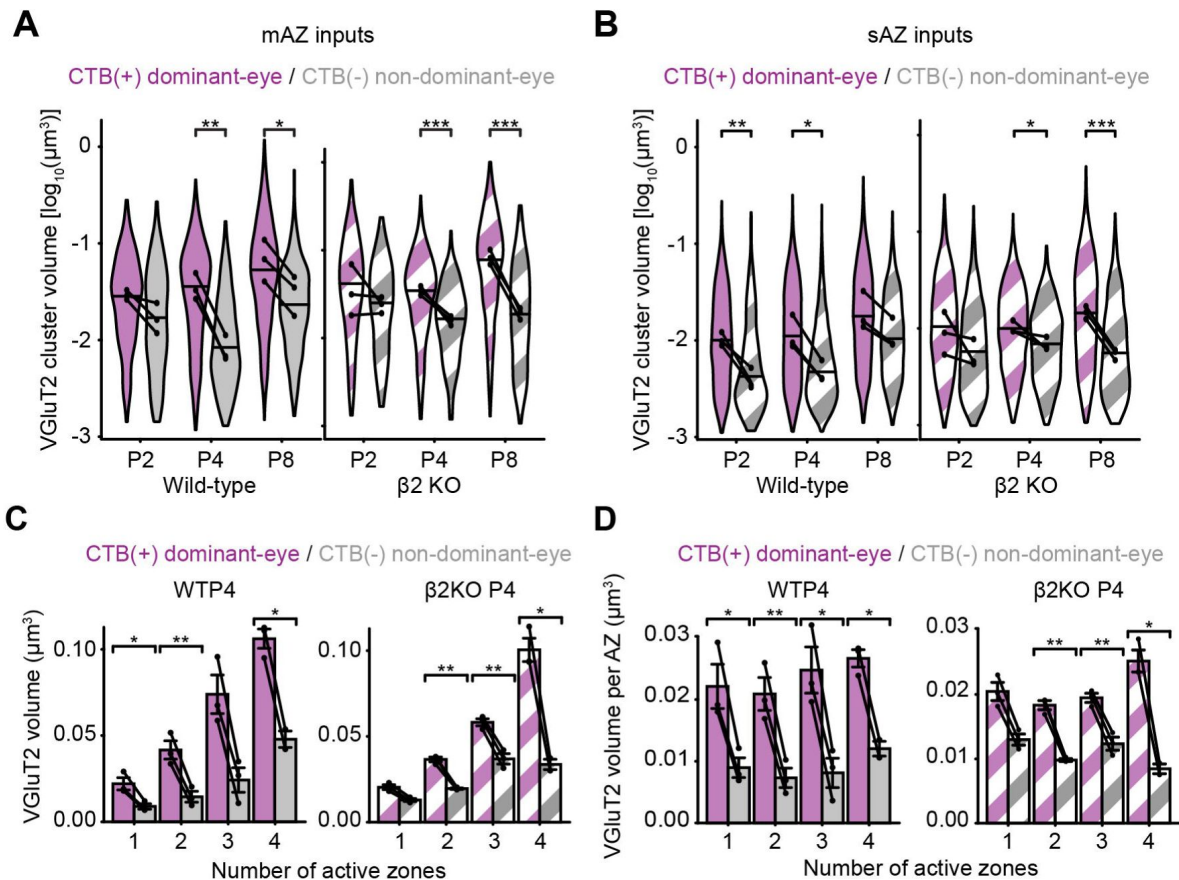


Fig. 3

Dominant-eye inputs show larger vesicle pools that scale with active zone number.

(A and B) Violin plots showing the distribution of VGlut2 cluster volume for (A) mAZ and (B) sAZ inputs in WT (filled) and $\beta 2$ KO mice (striped) at each age. The width of each violin plot reflects the relative synapse proportions across the entire grouped data set at each age ($N=3$ biological replicates) and the maximum width was normalized across all groups. The black dots represent the median value of each biological replicate ($N=3$), and the black horizontal lines represent the median value of all inputs grouped across replicates. Black lines connect measurements of CTB(+) and CTB(-) populations from the same biological replicate. Statistical significance was determined using a linear mixed model ANOVA with a post-hoc Bonferroni correction. Black asterisks indicate significant eye-specific differences at each age. * $p<0.05$, ** $p<0.01$, *** $p<0.001$. (C) Eye-specific VGlut2 signal volume for all inputs separated by number of AZs in WT (left panel) and $\beta 2$ KO mice (right panel) at P4. (D) Average VGlut2 volume per AZ for all inputs separated by number of AZs in WT (left panel) and $\beta 2$ KO mice (right panel) at P4. In panels (C) and (D), error bars indicate group means $\pm$ SEMs ($N=3$ biological replicates for each age and genotype). Black dots represent mean values from separate biological replicates and black lines connect eye-specific measurements within each replicate. Statistical significance between eye-specific measurements was assessed using two-tailed paired T-tests: * $p<0.05$; ** $p<0.01$.

we compared the Bassoon cluster number to VGlut2 volume for every retinogeniculate input. In both WT (Fig. S2D) and β 2KO mice (Fig. S2E), mAZ inputs contained an average of 2-3 Bassoon clusters (separate AZs) in the first postnatal week. In WT mice, CTB(+) dominant-eye mAZ inputs contained more AZs than CTB(-) non-dominant-eye mAZ inputs at P4 and P8 (Fig. S2D). This maturation was delayed until P8 in β 2KO mice (Fig. S2E). For CTB(+) dominant-eye inputs in both genotypes at P4, vesicle pool volume correlated positively with AZ number (Fig. 3C; Pearson correlation coefficients: WT [0.99] and β 2KO [0.95]). A similar, but weaker correlation was observed for CTB(-) non-dominant-eye inputs (Pearson correlation coefficients: WT [0.97] and β 2KO [0.90]). Dividing the total presynaptic VGlut2 volume by the AZ number revealed a consistent vesicle volume per AZ for both sAZ and mAZ inputs (Fig. 3D/S3, Table S3). Collectively, these results indicate that presynaptic vesicle pool volume scales with AZ number for each eye-specific input, while a dominant-eye bias persists throughout development.

Eye-specific synapse clustering before eye-opening

Eye-specific competition is thought to involve stabilization of co-active, neighboring inputs from the same eye and elimination of out of sync inputs from the opposite eye^{42–44}. Because axon refinement depends on the relative strength of neurotransmission between competing inputs³⁹,^{45–51}, mAZ inputs with multiple release sites could help stabilize nearby like-eye inputs⁴⁸,^{52–54}.

To quantify synaptic clustering patterns, we measured the distance from every eye-specific sAZ synapse to all other sAZ and mAZ inputs within each image field. sAZ synapses were often found nearby other inputs from the same eye (Fig. 4A). To define clustering, we searched volumetrically around each mAZ input and measured the fraction of sAZ synapses that were nearby at increasing distances (Fig. S4A/B). We then compared the observed percentages with datasets in which sAZ synapse positions were randomly shuffled within each neuropil volume. The largest differences occurred within a 1-2 μ m search distance (Fig. S4A/B), and so we chose a 1.5 μ m cutoff from the edge of each mAZ input to designate it “clustered” if at least one neighboring sAZ synapse lay within this distance and “isolated” if none did (Fig. 4B). No significant clustering was detected when sAZ and mAZ inputs originated from opposite eyes (Fig. S4C/D).

At the peak of competition (P4) in WT mice, more than 65% of both mAZ and sAZ inputs were clustered for both eyes, while this proportion fell to ~50% in β 2KO mice (Fig. 4B). In the WT dominant-eye projection, the fractions of clustered mAZ and sAZ inputs were similar at each age (Fig. 4C, top left panel). For the non-dominant eye projection, however, clustered mAZ inputs outnumbered clustered sAZ inputs at P4 (Fig. 4C, bottom left panel), the age when this eye adds sAZ synapses (Fig. 2C). β 2KO mice showed no difference between mAZ and sAZ clustering at any age (Fig. 4C, right panels). Additionally, in WT mice at P4 and P8, clustered mAZ inputs from both eyes had more neighboring sAZ synapses than did clustered sAZ synapses (Fig. 4D, left); this enrichment was absent in β 2KO mice (Fig. 4D, right). Thus, while most retinogeniculate synapses lie within 1.5 μ m of another like-eye input, WT mice show a relative increase in neighboring synapses near mAZ inputs during synaptic competition.

Clustered mAZ inputs in P4 WT mice were also closer together than isolated mAZ inputs. Clustered mAZ inputs formed by the dominant-eye were ~32% closer to the nearest like-eye clustered mAZ input compared to isolated mAZ inputs (Fig. 5A, left panel). In the non-dominant eye projection, clustered mAZ inputs were ~55% closer together compared to isolated mAZ inputs (Fig. 5B, left panel). Once segregation was complete at P8, distances between clustered and isolated mAZ inputs were more similar (Fig. S5). In β 2KO mice, distances between isolated and clustered mAZ inputs and their nearest clustered mAZ neighbor did not differ for either eye at P4 or P8 (Fig. 5C/S5, right panels). These patterns are consistent with sAZ synapse addition when mAZ inputs are in close proximity during competition.

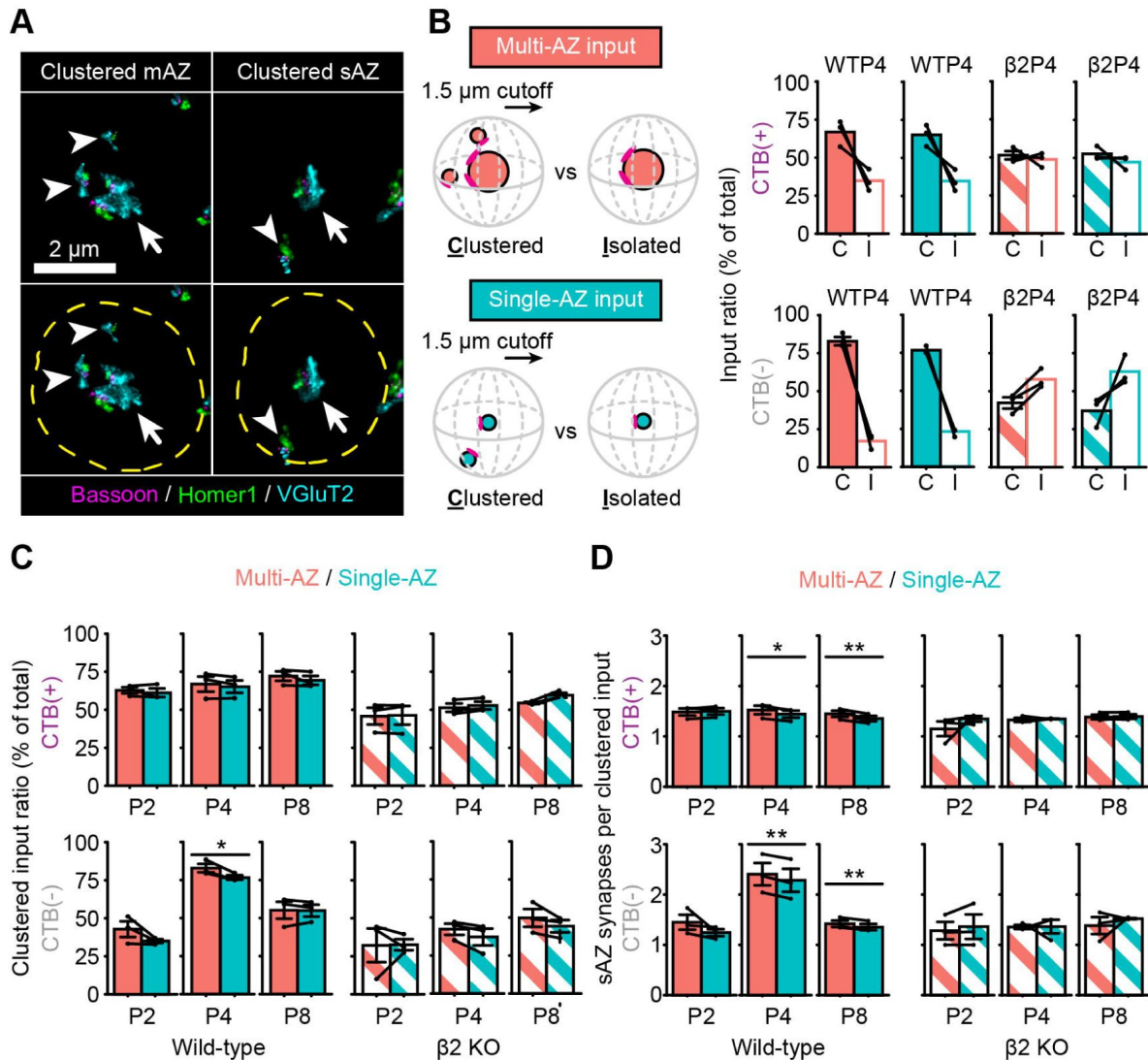


Fig. 4

Eye-specific synapse clustering before eye-opening.

(A) Representative mAZ (left panels) and sAZ (right panels) inputs in a WT P8 sample with nearby sAZ synapses (arrowheads) clustered within 1.5 μ m (dashed yellow ring). Arrows point to the centered mAZ or sAZ inputs. (B) Ratio of clustered and isolated mAZ and sAZ inputs for CTB(+) (upper panels) and CTB(-) (lower panels) inputs in WT and β 2KO mice at P4. (C) Comparison of the clustered input ratio between mAZ and sAZ inputs across different ages, genotypes, and eyes of origin. (D) Comparison of the average number of nearby sAZ synapses for clustered mAZ and sAZ inputs across different ages, genotypes, and eyes of origin. In panels B-D, black dots represent mean values from separate biological replicates and black lines connect measurements within each replicate (N=3 for each age and genotype). Error bars represent group means $\pm$ SEMs. Two-tailed paired T-tests were used to test statistical significance between mAZ and sAZ inputs at each age. * $p < 0.05$; ** $p < 0.01$.

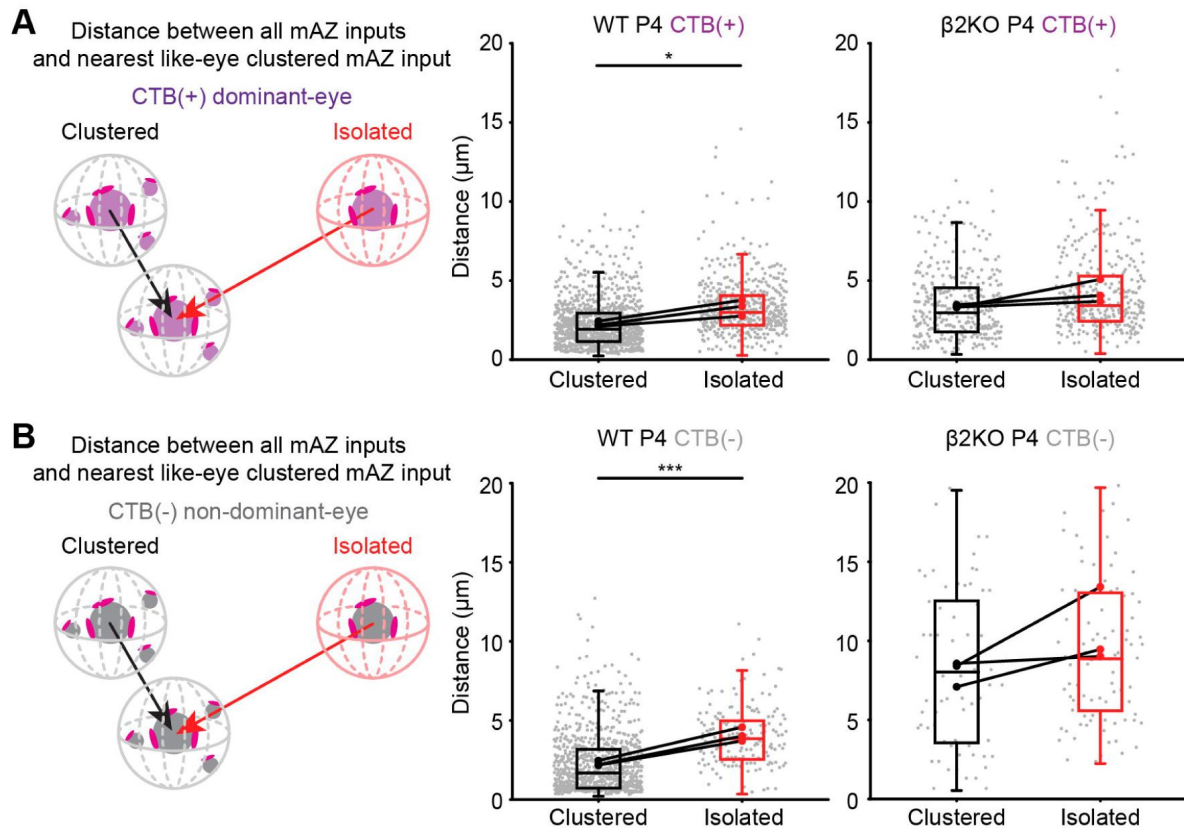


Fig. 5

Clustered mAZ inputs are closer than isolated inputs during competition.

(A-B) Distance between clustered and isolated mAZ inputs and the closest like-eye clustered mAZ input, shown for (A) CTB(+) and (B) CTB(-) projections at P4 in WT and β 2KO mice. Boxes indicate the 25%-75% distribution of input measurements from N=3 biological replicates and whiskers extend to 1.5 times the interquartile range. Gray dots represent individual distance measurements for all mAZ inputs. Black and red dots represent mean values from separate biological replicates and black lines connect measurements within each replicate (N=3 for each age and genotype). Statistical significance was determined using a linear mixed model ANOVA with post-hoc Bonferroni correction. Black asterisks indicate significant differences. * $p < 0.05$, *** $p < 0.001$.

Finally, we tested whether sAZ vesicle pool volume varies with distance to mAZ inputs. We classified each sAZ as “near” ($\leq 1.5 \mu\text{m}$) or “far” ($> 1.5 \mu\text{m}$) relative to the nearest like-eye mAZ input. Across ages and genotypes, vesicle pool volume was similar for near and far sAZ synapses in both eyes (Fig. S6). Thus, vesicle pool size in sAZ synapses appears independent of their proximity to mAZ inputs.

Discussion

Spontaneous retinal activity guides eye-specific refinement by strengthening dominant eye inputs in the correct territory and pruning weaker inputs from the competing eye. Using volumetric super-resolution microscopy and eye-specific synaptic immunolabeling, we previously identified activity-dependent, eye-specific differences in presynaptic vesicle pool size and vesicle association with the active zone (AZ), both favoring the dominant eye during synaptic competition²⁶. Here, we reanalyzed this dataset to measure synaptic spatial arrangements during normal development and how these are affected in a mutant model with abnormal eye-specific segregation.

Early input maturation and addition of release sites

Before eye-opening, retinal ganglion cell (RGC) axons from both eyes formed terminals with either multiple active zones (mAZ inputs) or a single active zone (sAZ synapses). Electron microscopy of dAPEX2-labeled ipsilateral RGC inputs revealed individual boutons with multiple active zones. Prior electron microscopy studies in mice showed no clustering from neighboring RGC boutons at postnatal day 8^{10, 11}, suggesting the mAZ inputs seen in STORM images are single boutons containing multiple release sites. During competition, the dominant eye projection generated more of these mAZ inputs, each with more active zones and larger vesicle pools than the non-dominant eye. We observed a linear relationship between presynaptic vesicle signal volume and active zone number, suggesting that release site addition and vesicle pool expansion may be linked during input maturation. Functional recordings indicate that release site addition is the primary driver of retinogeniculate input strengthening after eye-opening²⁴. Our findings reveal that eye-specific differences in release site addition at individual terminals emerge at the earliest stages of binocular refinement.

Spatial interactions during competition

Ipsilateral RGC axons are delayed in entering the dLGN until after contralateral inputs have already innervated the nucleus⁵⁵. Despite their late arrival, these axons competed by forming numerous sAZ inputs, which equalized the overall input density between the eyes in the future contralateral eye domain at postnatal day 4 (P4). Most of these synapses formed nearby other like-eye neighbors, and were enriched near mAZ inputs, supporting the idea of release site addition at strong synapses. Neighboring sAZ synapses could ultimately mature into mAZ inputs as release sites are added and vesicle pools expand.

Distance analysis showed that clustered mAZ boutons were closer together than isolated mAZ boutons, further supporting release site addition near strong neighboring inputs. It remains unknown, however, whether nearby mAZ inputs originate from the same RGC. Nearby mAZ inputs within individual RGC arbors could promote nearby release site addition via intrinsic mechanisms, such as increased presynaptic calcium entry or surface delivery of synaptogenic molecules. Boutons with multiple release sites could also trigger non-cell-autonomous signaling mechanisms that stabilize neighboring inputs from the same eye^{48, 53, 54}. Release site addition increases extracellular glutamate concentration and promotes spillover onto adjacent synapses, which enhances the excitability of developing relay cells²⁵ and could contribute to

long term synaptic plasticity evoked by high frequency spiking in retinal wave bursts ⁵⁶[56](#)–⁵⁹[59](#). Consequently, release site addition and local bouton clustering may act in tandem to stabilize coactive inputs.

Competitive refinement also involves synapse elimination and axonal retraction through punishment signals. Genetic deletions of VGluT2 or RIM1 proteins in ipsilaterally-projecting RGCs decreased presynaptic vesicle release and prevented retraction of contralateral RGC axons from the ipsilateral territory ³⁹[39](#), ⁴⁵[45](#). One downstream mediator of synaptic punishment is JAK2 kinase, which is phosphorylated in less active synapses ⁵²[52](#). Similar to neurotransmission mutant phenotypes, disruption of JAK2 signaling prevents axon retraction (punishment) during competition ⁵²[52](#). The spatial analyses we developed here will enable future mapping of input-specific punishment signals during synaptic competition. This includes phospho-JAK2 as well as molecular tags for glial pruning of weak inputs during eye-specific segregation ⁶⁰[60](#)–⁶²[62](#).

Requirement for spontaneous retinal activity

β 2KO mice showed significant defects in synapse addition during the first postnatal week. mAZs still formed with similar overall ratios as in wild type controls (Fig. S1, Table S3) and eye-specific differences in vesicle pool size still emerged. However, β 2KO mice failed to form many sAZ synapses at the height of competition, particularly in the late-arriving non-dominant (ipsilateral) eye projection. The failure to add synapses reduced synaptic clustering and more inputs formed in isolation in the mutants compared to controls.

While our findings support a role for spontaneous retinal activity in presynaptic release site addition and clustering, activity-dependent postsynaptic mechanisms also influence input refinement at later stages. Retinogeniculate synapses undergo postsynaptic strengthening and weakening through potentiation and depression mediated by AMPARs and NMDARs ⁵⁶[56](#)–⁵⁹[59](#). After eye-specific segregation, spontaneous retinal activity is required for postsynaptic AMPAR insertion, synaptic strengthening, and elimination of weaker inputs ⁶³[63](#). Continued maintenance of segregation depends on calcium influx into relay neurons via L-type calcium channels, further implicating postsynaptic signaling in late-stage refinement ⁶⁴[64](#), ⁶⁵[65](#). Input targeting may also be guided by molecular cues to form non-random, eye-specific connections with postsynaptic targets. Ipsilaterally-projecting RGCs have distinct gene expression profiles that specify axon guidance and may further support eye-specific synaptic targeting ⁶⁶[66](#), ⁶⁷[67](#). Spontaneous retinal activity may permit axons to read out molecular regulators of synaptogenesis, as previously shown for RGC axon retraction ⁶⁸[68](#).

Release site addition as a general mechanism underlying synaptic competition

Early synaptic clustering during retinogeniculate development resembles other circuits where neural activity guides competitive synaptic and axonal remodeling. At the neuromuscular junction (NMJ), motor neuron terminals compete for control of a postsynaptic muscle fiber; a single motor axon input strengthens while competing axons are eliminated ⁶⁹[69](#)–⁷¹[71](#). Competition at the NMJ depends on inter-synaptic distance, with motor axons losing connections located near stronger competing synapses ⁶⁹[69](#), ⁷⁰[70](#). As winning terminals are enlarged, presynaptic release site addition maintains a consistent density of Bassoon clusters ($\sim 2\text{--}3/\mu\text{m}^2$) ⁷²[72](#). Neighboring inputs with weaker synaptic transmission are selectively eliminated ⁷³[73](#), ⁷⁴[74](#). Competing motor axons differ in their release probabilities early in development ⁷⁵[75](#), suggesting that presynaptic efficacy triggers local “stabilization” signals in winners and “punishment” signals in losers ⁷⁶[76](#).

Presynaptic agrin release stabilizes postsynaptic receptor clusters by counteracting the destabilizing, dispersal effects of acetylcholine release, emphasizing the importance of early presynaptic transmission in postsynaptic stabilization ⁷⁷[77](#), ⁷⁸[78](#).

Similarly, in the developing cerebellum, Purkinje cells initially receive inputs from multiple climbing fibers (CFs) during the first postnatal week. Subsequently, one winning CF emerges and consolidates its synaptic inputs, while losing CFs are pruned ⁷⁹, ⁸⁰.

Here again, the winning input strengthens by adding presynaptic release sites, which increases multivesicular release and elevates glutamate concentration in the synaptic cleft ⁸¹–⁸³. Postsynaptic ultrastructural and molecular changes occur several days later as dominant CF inputs potentiate and losing CF inputs depress ⁸², ⁸⁴, ⁸⁵ through spike-timing dependent remodeling ⁸⁶, ⁸⁷. Across these models, the earliest distinguishing feature between competing inputs is relative presynaptic transmission strength. Thus, release-site addition may be a conserved mechanism that biases synaptic refinement outcomes across developing neural circuits.

Materials and methods

The raw imaging data in this paper were previously reported ²⁶. Materials and methods below are adapted from this work. All Matlab and Python code used in the work is available on GitHub (<https://github.com/SpeerLab>). Raw STORM images of the full data are available on the open-access Brain Imaging Library ⁸⁸. These images can be accessed here: <https://doi.org/10.35077/g.1164>.

Animals

Wild-type C57BL/6J mice (Stock Number 000664) and APEX2 mice (Stock Number 032764) used in this study were purchased from the Jackson Laboratory. ET33^{Cre} mice and β 2KO were generously gifted by Drs. Eric M. Ullian (University of California, San Francisco) and Michael C. Crair (Yale School of Medicine), respectively. All experimental procedures were performed in accordance with an animal study protocol approved by the Institutional Animal Care and Use Committee (IACUC) at the University of Maryland. Neonatal male and female mice were used interchangeably for all experiments. Tissue from biological replicates (N=3 animals) was collected for each age (P2/P4/P8) from each genotype (WT and β 2KO) (18 animals total). Primers used for genotyping β 2KO mice are: forward: CAGGCGTTATCCACAAAGACAGA; reverse:

TTGAGGGGAGCAGAACAGAATC; mutant reverse: ACTTGGGTTTGGGCGTGTGAG.

Primers used for genotyping ET33Cre mice are: Forward: GGTCTTGGCAGATGGGCAT; reverse: CGGCAAACGGACAGAAGCATT.

Electron microscopy tissue preparation

Animals were deeply anesthetized with ketamine/xylazine and transcardially perfused with 5–10 mls of 37°C 0.9% sterile saline (pH 7.2) followed by 20–30 mls of 37°C 4% EM-Grade paraformaldehyde (PFA, Electron Microscopy Sciences), 2% EM-Grade glutaraldehyde (GA, Electron Microscopy Sciences), 4mM calcium chloride (CaCl₂, Sigma-Aldrich) in 0.2M cacodylate buffer (pH 7.4). Brains were postfixed in the same perfusion fixative solution overnight at 4°C. Brains were vibratome sectioned in 0.2M cacodylate buffer (pH 7.4) at 100 μ m. A circular tissue punch (~500 μ m diameter) containing the dLGN was microdissected from each section using a blunt-end needle. dLGN sections were washed in 0.2M cacodylate buffer (pH 7.4) at 4°C on a rotator (4x 20 minutes each). Sections were incubated in 20mM glycine (Sigma-Aldrich) in 0.2M cacodylate buffer (pH 7.4) at 4°C on a rotator for 30 minutes, followed by 5 x 20 minutes washes in 0.2M cacodylate buffer (pH 7.4) at 4°C on a rotator. dLGN sections were immersed in 0.5mg/mL DAB solution, covered with foil, for 30 minutes at 4°C on a rotator. 10 μ L of 0.03% hydrogen

peroxide (Sigma-Aldrich) was mixed into the DAB solution (1 mL) and incubated for 10 minutes at 4°C on a rotator (light protected). The reaction was quenched with 0.2M cacodylate buffer (pH 7.4) washes (3 x 1-minute washes followed by 2 x 20 minutes washes).

Tissue preparation for scanning electron microscopy

dLGN sections were fixed in 2% osmium tetroxide (Electron Microscopy Sciences) in 0.15M cacodylate buffer (pH 7.4) for 45 minutes at room temperature. Sections were reduced with 2.5% potassium ferricyanide (Electron Microscopy Sciences) in 0.15M cacodylate buffer (pH 7.4) for 45 minutes at room temperature in the dark, followed by 2 x 10-minute washes in double distilled water. Sections were incubated in 1% aqueous thiocarbonylhydrazide (Electron Microscopy Sciences) for 20 minutes, followed by 2 x 10-minute washes in double distilled water. Samples were fixed in 1% aqueous osmium tetroxide for 45 minutes at room temperature, followed by 2 x 10-minute washes in double distilled water. Sections were postfixed in 1% uranyl acetate (Electron Microscopy Sciences) in 25% ethanol in the dark for 20 minutes at room temperature and washed and stored in double distilled water overnight. The next day, samples were stained with aqueous lead aspartate at 60°C for 30 minutes and washed with double distilled water 2 x 10 minutes. Tissues were dehydrated in a graded dilution series of 100% ethanol (35%; 50%; 70%; 95%; 100%; 100%; 100% EtOH) for 10 minutes each at room temperature. Samples were immersed in propylene oxide (Electron Microscopy Sciences) 3 x 10 minutes at room temperature, followed by a series of epon resin/propylene oxide (812 Epon Resin, Electron Microscopy Sciences) exchanges with increasing resin concentrations (50% resin/50% propylene oxide; 65% resin/35% propylene oxide; 75% resin/25% propylene oxide; 100% resin; 100% resin) for 90 minutes each. Tissues were transferred to BEEM capsules (Electron Microscopy Sciences) that were filled with 100% resin and polymerized for at least 48 hours at 60°C.

Transmission electron microscopy image acquisition

Plasticized sections were cut at 70 nm with a Histo Jumbo diamond knife (DiATOME) using a Leica UC7 ultramicrotome. Sections were decompressed using chloroform vapor and collected onto 3.05mm 200 mesh Gilder thin bar hexagonal mesh copper grids (T200H-Cu, Electron Microscopy Sciences). Sections were imaged unstained on a Hitachi HT7700 transmission electron microscope (HT7700, Hitachi High-Tech America, Inc.) at 80kV.

Eye injections

Intraocular eye injections were performed one day before tissue collection. Briefly, mice were anesthetized by inhalant isoflurane and sterile surgical spring scissors were used to gently part the eyelid to expose the corneoscleral junction. A small hole was made in the eye using a sterile 34-gauge needle and ~0.5 µL of cholera toxin subunit B conjugated with Alexa Fluor 488 (CTB-488, ThermoFisher Scientific, Catalogue Number: C34775) diluted in 0.9% sterile saline was intravitreally pressure-injected into the right eye using a pulled-glass micropipette coupled to a Picospritzer (Parker Hannifin).

dLGN tissue preparation for STORM imaging

Animals were deeply anesthetized with ketamine/xylazine and transcardially perfused with 5-10 mls of 37°C 0.9% sterile saline followed by 10 mls of room temperature 4% EM-Grade paraformaldehyde (PFA, Electron Microscopy Sciences) in 0.9% saline.

Brains were embedded in 2.5% agarose and vibratome sectioned in the coronal plane at 100 µm. From the full anterior-posterior series of dLGN sections (~6-8 sections) we selected the central two sections for staining in all biological replicates. These sections were morphologically consistent with Figs. 134-136 (5.07-5.31 mm) of the postnatal day 6 mouse brain from Paxinos's "Atlas of the developing mouse brain" Academic Press, 2020 ⁸⁹[89](#). Selected sections were postfixed in 4% PFA for 30 minutes at room temperature and washed for 30-40 minutes in 1X PBS. The dLGN was

identified by the presence of CTB-488 signals using a fluorescence dissecting microscope. A circular tissue punch (~500 μm diameter) containing the dLGN was microdissected from each section using a blunt-end needle. A small microknife cut was made at the dorsal edge of the dLGN which, together with the CTB-488 signal, enabled us to identify the dLGN orientation during image acquisition.

Immunohistochemistry

We used a serial-section single-molecule localization imaging approach to prepare samples and collect super-resolution fluorescence imaging volumes as previously described²⁶. dLGN tissue punches were blocked in 10% normal donkey serum (Jackson ImmunoResearch, Catalogue Number: 017-000-121) with 0.3% Triton X-100 (Sigma-Aldrich Inc.) and 0.02% sodium azide (Sigma-Aldrich Inc.) diluted in 1X PBS for 2-3 hours at room temperature and then incubated in primary antibodies for ~72 hours at 4°C. Primary antibodies used were Rabbit anti-Homer1 (Synaptic Systems, Catalogue Number: 160003, 1:100) to label postsynaptic densities (PSDs), mouse anti-Bassoon (Abcam, Catalogue Number AB82958, 1:100) to label presynaptic active zones (AZs), and guinea pig anti-VGluT2 (Millipore, Catalogue Number AB251-I, 1:100) to label presynaptic vesicles. Following primary antibody incubation, tissues were washed in 1X PBS for 6 x 20 minutes at room temperature and incubated in secondary antibody solution overnight for ~36 hours at 4°C. The secondary antibodies used were donkey anti-rabbit IgG (Jackson ImmunoResearch, Catalogue Number 711-005-152, 1:100) conjugated with Dy749P1 (Dyomics, Catalogue Number 749P1-01) and Alexa Fluor 405 (ThermoFisher, Catalogue Number: A30000), donkey anti-mouse IgG (Jackson ImmunoResearch, Catalogue Number 715-005-150, 1:100) conjugated with Alexa Fluor 647 (ThermoFisher, Catalogue Number: A20006) and Alexa Fluor 405, and donkey anti-guinea pig IgG (Jackson ImmunoResearch, Catalogue Number 706-005-148, 1:100) conjugated with Cy3b (Cytiva, Catalogue Number: PA63101). Tissues were washed 6 x 20 minutes in 1X PBS at room temperature after secondary antibody incubation.

Postfixation, dehydration, and embedding in epoxy resin

Tissue embedding was performed as previously described²⁶. Tissues were postfixated with 3% PFA + 0.1% GA (Electron Microscopy Sciences) in PBS for 2 hours at room temperature and then washed in 1X PBS for 20 minutes. To plasticize the tissues for ultrasectioning, the tissues were first dehydrated in a graded dilution series of 100% ethanol (50%/70%/90%/100%/100% EtOH) for 15 minutes each at room temperature and then immersed in a series of epoxy resin/100% EtOH exchanges (Electron Microscopy Sciences) with increasing resin concentration (25% resin/75% ethanol; 50% resin/50% ethanol; 75% resin/25% ethanol; 100% resin; 100% resin) for 2 hours each. Tissues were transferred to BEEM capsules (Electron Microscopy Sciences) that were filled with 100% resin and polymerized for 16 hours at 70°C.

Ultrasectioning

Plasticized tissue sections were cut using a Leica UC7 ultramicrotome at 70 nm using a Histo Jumbo diamond knife (DiATOME). Chloroform vapor was used to reduce compression after cutting. For each sample, ~100 sections were collected on a coverslip coated with 0.5% gelatin and 0.05% chromium potassium (Sigma-Aldrich Inc.), dried at 60 degrees for 25 minutes, and protected from light prior to imaging.

Imaging chamber preparation

Coverslips were chemically etched in 10% sodium ethoxide for 5 minutes at room temperature to remove the epoxy resin and expose the dyes to the imaging buffer for optimal photoswitching. Coverslips were then rinsed with ethanol and dH₂O. To create fiducial beads for flat-field and chromatic corrections, we mixed 715/755nm and 540/560nm, carboxylate-modified microspheres (Invitrogen, Catalogue Numbers F8799 and F8809, 1:8 ratio respectively) to create a high-density fiducial marker and then further diluted the mixture at 1:750 with Dulbecco's PBS to create a low-

density bead solution. Both high- and low-density bead solutions were spotted on the coverslip ($\sim 0.7 \mu\text{L}$ each) for flat-field and chromatic aberration correction respectively. Excess beads were rinsed away with dH₂O for 1–2 minutes. The coverslip was attached to a glass slide with double-sided tape to form an imaging chamber. The chamber was filled with STORM imaging buffer (10% glucose, 17.5 μM glucose oxidase, 708 nM catalase, 10 mM MEA, 10 mM NaCl, and 200 mM Tris) and sealed with epoxy.

Imaging setup

Imaging was performed using a custom single-molecule super-resolution imaging system. The microscope contained low (4x/10x air) and high (60x 1.4NA oil immersion) magnification objectives mounted on a commercial frame (Nikon Ti-U) with back optics arranged for oblique incident angle illumination. We used continuous-wave lasers at 488 nm (Coherent), 561 nm (MPB), 647 nm (MPB), and 750 nm (MPB) to excite Alexa 488, Cy3B, Alexa 647, and Dy749P1 dyes respectively. A 405 nm cube laser (Coherent) was used to reactivate Dy749P1 and Alexa647 dye photoswitching. The microscope was fitted with a custom pentaband/pentanotch dichroic filter set and a motorized emission filter wheel. The microscope also contained an IR laser-based focus lock system to maintain optimal focus during automatic image acquisition. Images were collected on 640 $\times$ 640-pixel region of an sCMOS camera (ORCA-Flash4.0 V3, Hamamatsu Photonics) with a pixel size of ~ 155 nm.

Automated image acquisition

Fiducials and tissue sections on the coverslip were imaged using the low magnification objective (4X) to create a mosaic overview of the specimen. Beads/sections were then imaged at high magnification (60X) to select regions of interest (ROIs) in the Cy3B and Alexa 488 channels. Before final image acquisition, laser intensities and the incident angle were adjusted to optimize photoswitching for STORM imaging and utilize the full dynamic range of the camera for conventional imaging.

Low-density bead images were taken in 16 partially overlapping ROIs. 715/755 nm beads were excited using 750 nm light and images were collected through Dy749P1 and Alexa 647 emission filters. 540/560 nm beads were excited using a 488 nm laser and images were collected through Alexa 647, Cy3B, and Alexa 488 emission filters. These fiducial images were later used to generate a non-linear warping transform to correct chromatic aberration. Next, ROIs within each tissue section were imaged at conventional (diffraction-limited) resolution in all four-color channels sequentially.

Following conventional image acquisition, a partially overlapping series of 9 images were collected in the high-density bead field for all 4 channels (Dy749P1, Alexa 647, Cy3B, and Alexa 488). These images were later used to perform a flat-field image correction of non-uniform laser illumination across the ROIs. Another round of bead images was taken as described above in a different ROI of the low-density bead field. These images were later used to confirm the stability of chromatic offsets during imaging. All ROIs within physical sections were then imaged by STORM for Dy749P1 and Alexa 647 channels. Images were acquired using a custom progression of increasing 405 nm laser intensity to control single-molecule switching. 8000 frames of Dy749P1 channel images were collected (60 Hz imaging) followed by 12000 frames of Alexa 647 channel images (100 Hz). In a second imaging pass, the same ROIs were imaged for Cy3B and Alexa 488 channels, each for 8000 frames (60 Hz).

We imaged the ipsilateral and contralateral ROIs separately in each physical section of the dLGN. For consistency of ROI selection across biological replicates at each age, we identified the dorsal-ventral (DV) axis of the dLGN and selected ROIs within the center (core region) at 2/5 (ipsilateral ROI) and 4/5 (contralateral ROI) of the full DV length.

Image processing

Single-molecule localization was performed using a previously described DAOSTORM algorithm modified for use with sCMOS cameras^{90,91}. Molecule lists were rendered as 8-bit images with 15.5 nm pixel size where each molecule is plotted as an intensity distribution with an area reflecting its localization precision. Low-density fiducial images were used for chromatic aberration correction. We localized 715/755 beads in Dy749P1 and Alexa 647 channels, and 540/560 beads in Alexa 647, Cy3B, and Alexa 488 channels. A third-order polynomial transform map was generated by matching the positions of each bead in all channels to the Alexa 647 channel. The average residual error of bead matching was <15 nm for all channels. The transform maps were applied to both 4-color conventional and STORM images. Conventional images were upscaled (by 10X) to match the STORM image size. The method to align serial sections was previously described²⁶. STORM images were first aligned to their corresponding conventional images by image correlation. To generate an aligned 3D image stack from serial sections, we normalized the intensity of all Alexa 488 images and used these normalized images to generate both rigid and elastic transformation matrices for all four-color channels of both STORM and conventional data. The final image stack was then rotated and cropped to exclude incompletely imaged edge areas. To further confirm that the processed region corresponds to the contralateral dLGN, conventional CTB(+) signals of the labeled contralateral projection were thresholded to create a polygonal mask (a convex hull analysis linking the outermost CTB signals in the image volume).

The mask was then applied to STORM images to exclude peripheral areas where CTB signals were absent or faint.

Cell body filter

The aligned STORM images had non-specific labeling of cell bodies in Dy749P1 and Alexa 647 channels corresponding to Homer1 and Bassoon immunolabels. To limit synaptic cluster identification to the neuropil region we identified cell bodies based on their Dy749P1 signal and excluded these regions from further image processing.

STORM images were convolved with a Gaussian function ($\sigma=140$ nm) and then binarized using the lower threshold of a two-level Otsu threshold method. We located connected components in the thresholded images and generated a mask based on components larger than e^{11} voxels. Because cell body clusters were orders of magnitude larger than synaptic clusters, the cell body filter algorithm was robust to a range of size thresholds. The mask was applied to images of all channels to exclude cell body areas.

Eye-specific synapse identification and quantification

To correct for minor variance in image intensity across physical sections, we normalized the pixel intensity histogram of each section to the average histogram of all sections.

Image histograms were rescaled to make full use of the 8-bit range. Using a two-level Otsu threshold method, the conventional images were thresholded into three classes: a low-intensity background, low-intensity signals above the background representing non-synaptic labeling, and high-intensity signals representing synaptic structures. The conventional images were binarized by the lower two-level Otsu threshold, generating a mask for STORM images to filter out background signals. STORM images were convolved with a Gaussian function ($\sigma=77.5$ nm) and thresholded using the higher two-level Otsu threshold. Following thresholding, connected components were identified in three dimensions using MATLAB 'conncomp' function. A watershedding approach was applied to split large clusters that were improperly connected.

Clusters were kept for further analysis only if they contained aligned image information across two or more physical sections. We also removed all edge synapses from our analysis by excluding synapses that did not have blank image data on all adjacent sides.

To distinguish non-specific immunolabeling from true synaptic signals, we quantified two parameters for each cluster: cluster volume and cluster signal density calculated by the ratio of within-cluster pixels with positive signal intensity in the raw STORM images.

Two separate populations were identified in 2D histograms plotted from these two parameters. We manually selected the population with higher volumes and signal densities representing synaptic structures. To test the robustness of the manual selection, we performed multiple repeated measurements of the same data and discovered a between-measurement variance of <1% (data not shown).

To identify paired pre-and postsynaptic clusters, we first measured the centroid-centroid distance of each cluster in the Dy749P1 (Homer1) and Alexa 647 (Bassoon) channels to the closest cluster in the other channel. We next quantified the signal intensity of each opposing synaptic channel within a 140 nm shell surrounding each cluster. A 2D histogram was plotted based on the measured centroid-centroid distances and opposing channel signal densities of each cluster. Paired clusters with closely positioned centroids and high intensities of apposed channel signal were identified using the OPTICS algorithm. Retinogeniculate synapses were identified by pairing Bassoon (Alexa 647) clusters with VGluT2 (Cy3B) clusters using the same method as pre/post-synaptic pairing. Synapses from the right eye were identified by pairing VGluT2 clusters with CTB (Alexa 488) clusters. The volume of each cluster reflected the total voxel volume of all connected voxels, and the total signal intensity was a sum of voxel intensity within the volume of the connected voxels.

Multi-AZ synapse identification and quantification

To determine whether an eye-specific VGluT2 cluster is a mAZ synapse or a sAZ synapse, we measured the number of active zones (defined by individual Bassoon clusters) associated with each VGluT2 cluster in the dataset. A 3D shell was extended 140 nm from the surface voxels of each VGluT2 cluster and any Bassoon clusters that fell within the shell were considered to be associated with the target VGluT2 cluster.

The number of active zones (AZs) associated with each VGluT2 cluster was then measured. VGluT2 clusters associated with more than 1 AZ were defined as mAZ synapses, while those associated with only 1 AZ were defined as sAZ synapses.

Quantification of mAZ and sAZ synapse VGluT2 cluster volume was performed using the “regionprops” function in MATLAB, which provided the voxel size and weighted centroid of each VGluT2 cluster. The search for sAZ synapses adjacent to mAZ synapses (synaptic clustering analysis) was conducted using a similar search approach as for associated Bassoon clusters, with expansion shell sizes ranging from 1 μm to 4 μm from the surface voxels of each mAZ synapse. The main Figs. in the study utilized an expansion distance of 1.5 μm . An eye-specific sAZ synapse was considered to be near a mAZ synapse if its weighted centroid fell within the expanded volume.

Quantification and statistical analysis

Statistical analysis was performed using SPSS. Plots were generated by SPSS or R (ggplot2). Statistical details are presented in the figure legends and supplemental tables S2/S3. For all measurements in this paper, we analyzed $N = 3$ biological replicates (individual mice) for each genotype (WT and $\beta 2\text{KO}$) at each age (P2, P4, and P8).

Cluster densities, synapse AZ number, average VGluT2 cluster volume, and all fractional measurements were presented as mean $\pm$ SEM values in paired bar graphs and statistical analysis was performed using two-tailed paired-T tests. Nonparametric Kolmogorov-Smirnov tests were used in all cumulative histogram comparisons. We used a linear mixed model to compare VGluT2 cluster volumes (**Figs. 3** and S2) and distance measurements (**Figs. 5** and S5). For VGluT2 cluster volume comparisons, the age or eye-of-origin was the fixed main factor and biological replicate IDs were nested random factors. In distance measurement comparisons, the mAZ synapse AZ number was the fixed main factor and biological replicate IDs were nested random factors. In violin plots, each violin showed the distribution of grouped data from all biological replicates from the same condition. Each black dot represents the median value of each biological replicate, and the horizontal black line represents the group median. Black lines connect measurements of CTB(+) and CTB(-) populations from the same biological replicate. Asterisks in all figures indicate statistical significance: * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$. For VGluT2 volume comparisons, we calculated 5/95% confidence intervals based on the linear mixed model. 5/95% confidence intervals in synapse densities, AZ numbers, and distances were calculated for paired or unpaired data comparisons using SPSS.

All statistical results are listed in Table S2. All conclusions we draw in the main text from the data have corresponding confidence intervals listed in both the figure legends and Table S3.

Acknowledgements

We thank Drs. Michael C. Crair (Yale University) and Eric M. Ullian (University of California, San Francisco) for generously sharing the $\beta 2\text{KO}$ and ET33^{Cre} mouse lines used in this work. We are grateful to Tim Maugel in the Laboratory for Biological Ultrastructure (UMD) for his assistance with transmission electron microscopy experiments. The research was funded by National Institutes of Health grant DP2MH125812 (C.M.S.).

Additional information

Author contributions

Conceptualization, C.Z. and C.M.S.; data curation, C.Z., T.V., C.M.S.; formal analysis, C.Z. and C.M.S.; funding acquisition, C.M.S.; investigation, C.Z., T.V., C.M.S.; methodology, C.Z., T.V., C.M.S.; project administration, C.Z. and C.M.S.; resources, C.Z., T.V., C.M.S.; software, C.Z. and C.M.S.; supervision, C.M.S.; validation, C.Z., T.V., C.M.S.; visualization, C.Z., T.V., C.M.S.; writing – original draft preparation, C.Z. and C.M.S.; writing – review & editing, C.Z., T.V., C.M.S.

Funding

National Institutes of Health (DP2MH125812)

Additional files

Supplemental Table S1 [↗](#)

Supplemental Table S2 [↗](#)

Supplemental Table S3 [↗](#)

CTB(+) dominant-eye
CTB(-) non-dominant-eye

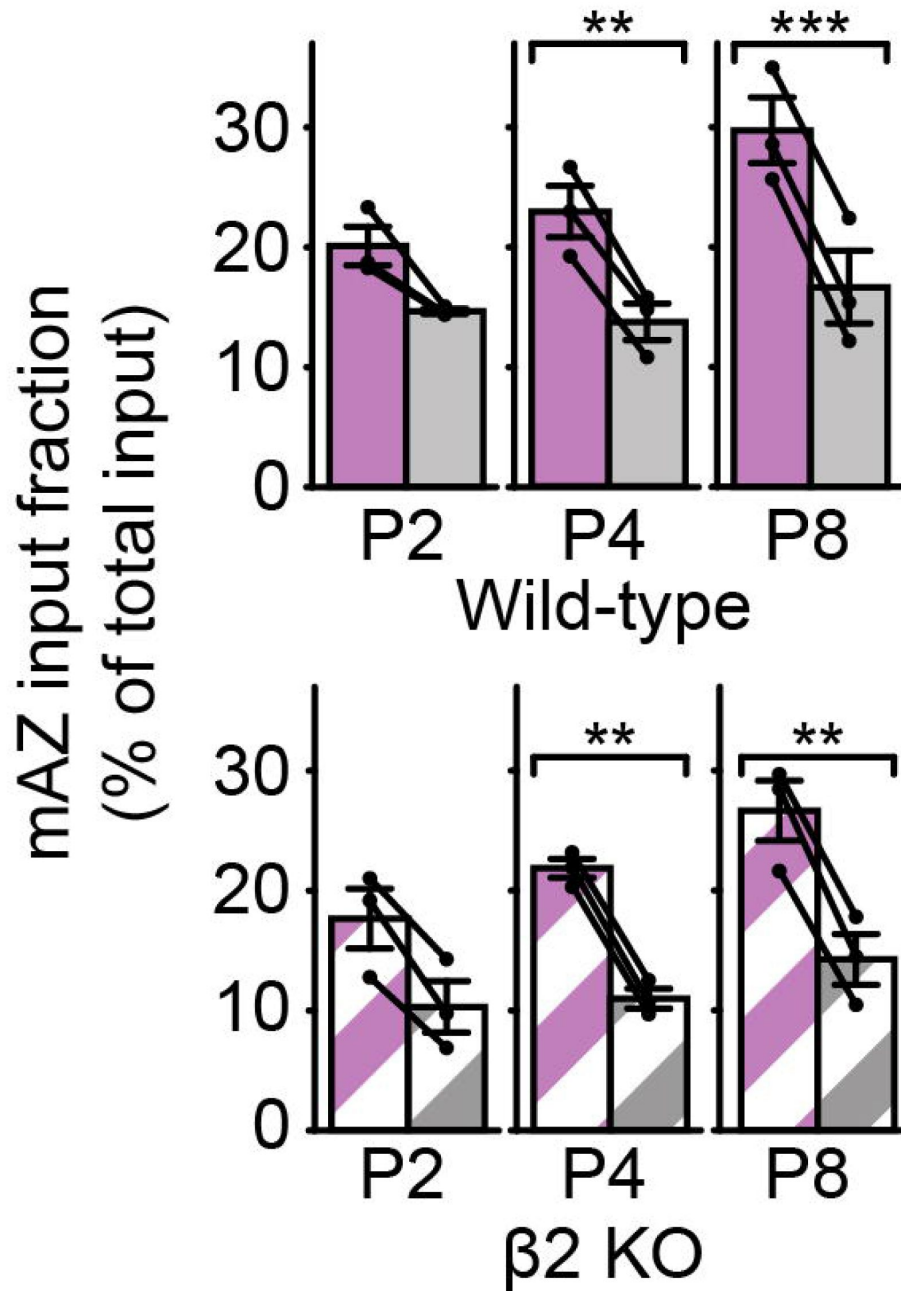


Fig. S1

Fraction of mAZ inputs across development, related to Fig. 2.

Eye-specific mAZ input fraction across development in WT (top panel) and $\beta 2$ KO mice (bottom panel). Black dots represent mean values from separate biological replicates and black lines connect measurements within each replicate (N=3 for each age and genotype). Error bars represent group means $\pm$ SEMs. Statistical significance between eye-specific measurements was assessed using two-tailed paired T-tests. **p<0.01, ***p<0.001

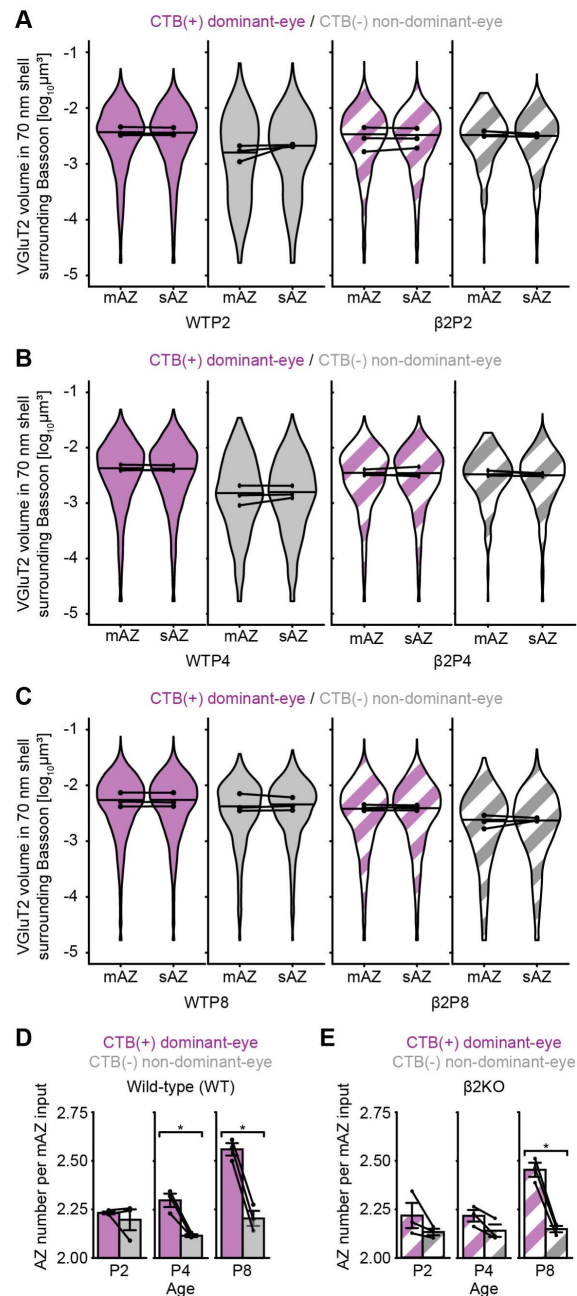


Fig. S2

Quantification of docked vesicle pool volume and AZ number in mAZ and sAZ inputs, related to Fig. 2

(A-C) Violin plots show the distribution of VGlut2 cluster volume within a 70 nm shell surround individual Bassoon clusters at (A) P2, (B) P4, and (C) P8. The width of each violin plot reflects the relative synapse proportions at each volume across the entire grouped data set ($N=3$ biological replicates). The maximum width of the violin plots was normalized across all groups. Black horizontal lines represent the median value of all inputs grouped across replicates. The black dots represent the median value of each biological replicate and black lines between dots connect measurements from the same biological replicate. Statistical significance was determined using a linear mixed model ANOVA with post-hoc Bonferroni correction. See Table S3 for 5/95% confidence intervals. (D-E) Average AZ number per mAZ input in (D) WT and (E) β 2KO mice. Black dots represent mean values from separate biological replicates ($N = 3$) and black lines connect measurements within each replicate. Error bars represent means $\pm$ SEMs. Statistical significance was assessed using two-tailed paired T-tests. * $p<0.05$.

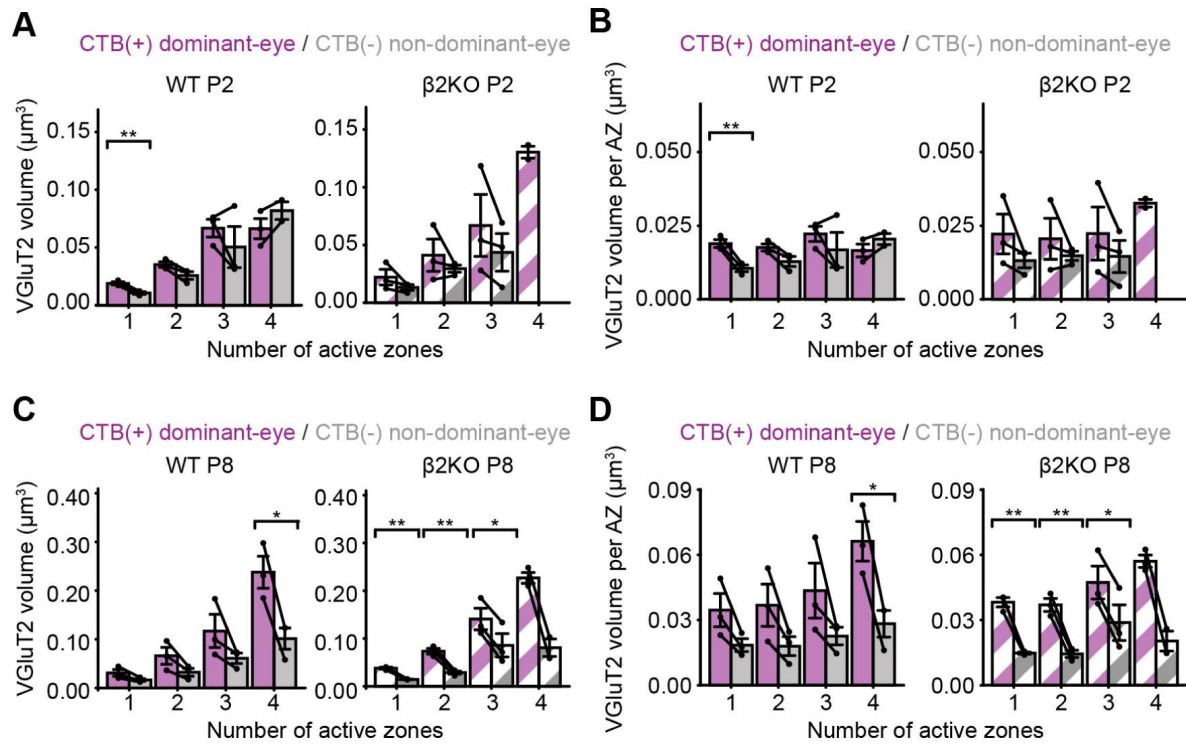


Fig. S3

Relationship between vesicle pool volume and active zone number, related to Fig. 3

(A) Total VGlut2 volume per input and (B) average VGlut2 volume per AZ for all inputs separated by number of AZs in WT (filled bars) and β 2KO mice (striped bars) at P2. (C) Total VGlut2 volume per input and (D) average VGlut2 volume per AZ for all inputs separated by number of AZs in WT (filled bars) and β 2KO mice (striped bars) at P8. In all panels, error bars indicate group means $\pm$ SEMs (N=3 biological replicates for each age and genotype). Black dots represent mean values from separate biological replicates and black lines connect measurements within each replicate. Statistical significance between eye-specific measurements was assessed using two-tailed paired T-tests: * $p < 0.05$; ** $p < 0.01$.

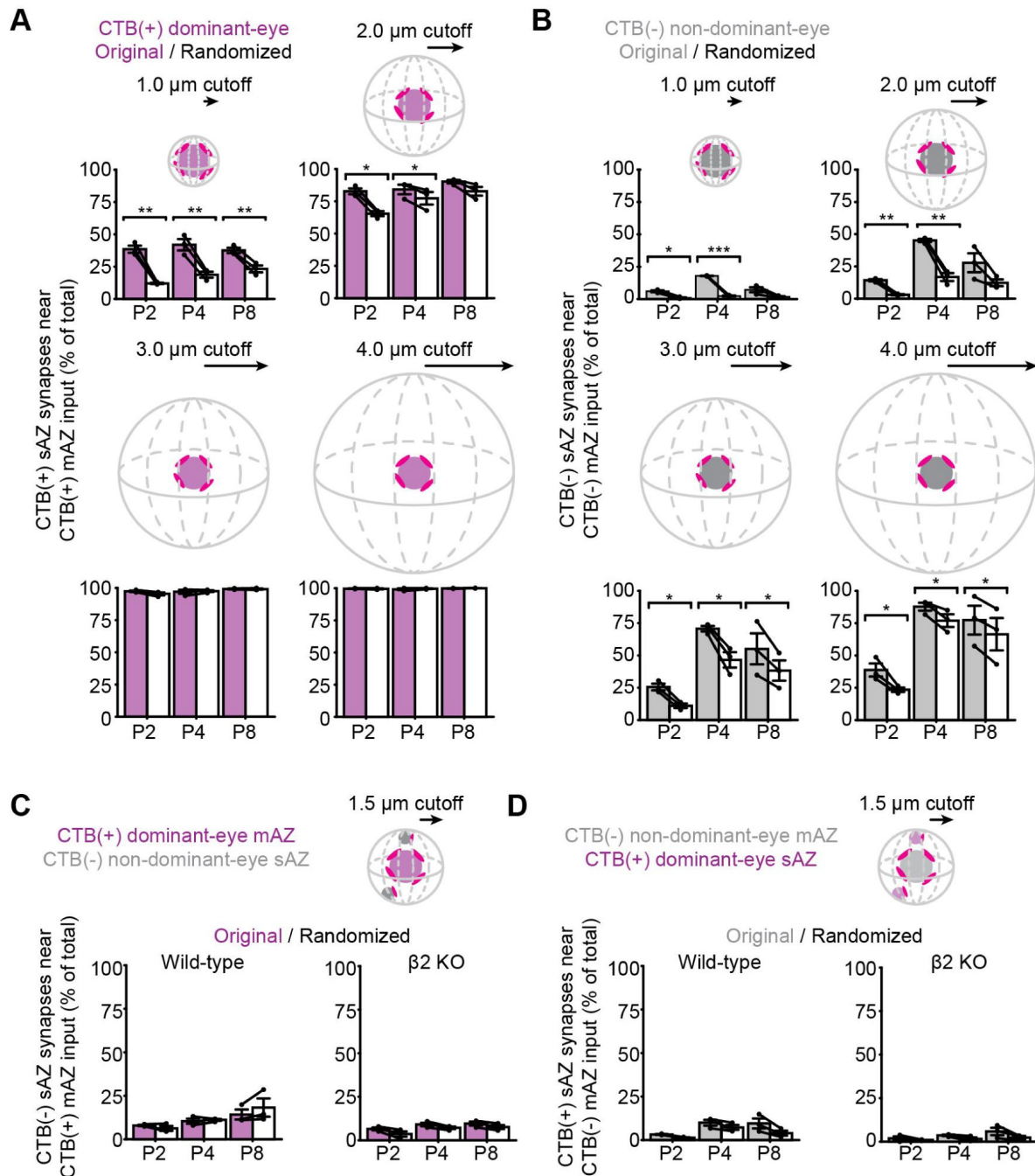


Fig. S4

sAZ synapse clustering near like-eye mAZ inputs, related to Fig. 4

(A) The ratio of CTB(+) dominant-eye and (B) CTB(-) non-dominant-eye sAZ synapses nearby like-eye mAZ inputs within increasing distance cutoffs in WT samples across ages. At distances of 1-2 μm across all ages, the observed ratios (filled bars) are higher than a reshuffling of the data (open bars) where the position of sAZ inputs was randomized within the neuropil. (C-D) Percentage of sAZ synapses within 1.5 μm of opposite-eye mAZ inputs across ages in WT and $\beta 2$ KO mice. Data are compared against a randomization of sAZ positions (open bars) as in panels A/B. For all panels, error bars represent group means $\pm$ SEMs (N=3 biological replicates for each age and genotype). Black dots represent mean values from separate biological replicates and black lines connect measurements within each replicate. Statistical significance was assessed using two-tailed paired T-tests: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

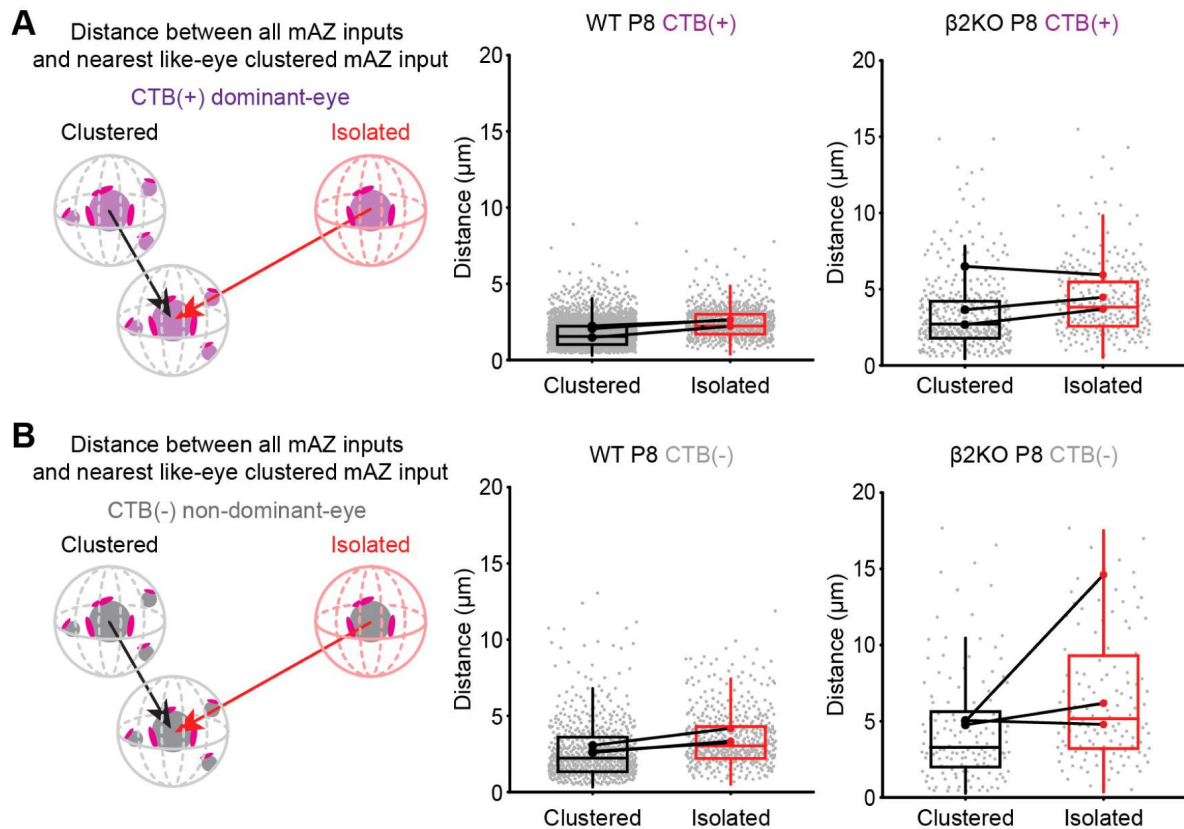


Fig. S5

Clustered and isolated mAZ inputs show similar spacing after competition, related to Fig. 5

(A-B) Distance between clustered and isolated mAZ inputs and the closest like-eye clustered mAZ input, shown for (A) CTB(+) and (B) CTB(-) projections at P8 in WT and $\beta 2\text{KO}$ mice. Boxes indicate the 25%-75% distribution of input measurements from $N=3$ biological replicates and whiskers extend to 1.5 times the interquartile range. Gray dots represent individual distance measurements for all mAZ inputs. Black and red dots represent mean values from separate biological replicates and black lines connect measurements within each replicate ($N=3$ for each age and genotype). Statistical significance was determined using a linear mixed model ANOVA with post-hoc Bonferroni correction (see Table S3 for 5/95% confidence intervals).

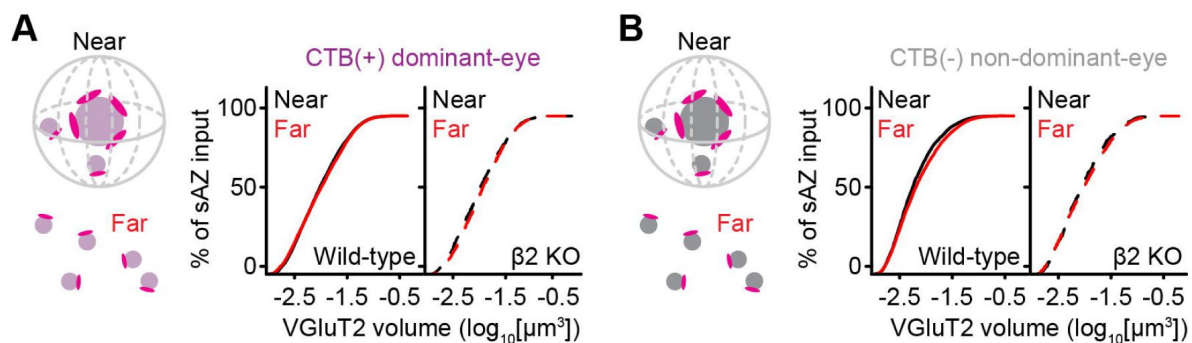


Fig. S6

sAZ synapse vesicle pool volume is independent of distance to mAZ inputs, related to Fig. 5

(A) Cumulative distributions of VGLUT2 volume for CTB(+) dominant-eye or (B) CTB(-) non-dominant-eye sAZ synapses near ($<1.5 \mu\text{m}$) or far from ($>1.5 \mu\text{m}$) like-eye mAZ inputs. The distributions show merged data across all ages (P2/P4/P8; $N=3$ biological replicates at each time point). A nonparametric Kolmogorov-Smirnov test was used for statistical analysis (see Table S3 for 5/95% confidence intervals).

References

1. Rall W (1962) **Electrophysiology of a dendritic neuron model** *Biophys J* **2**:145–67 [https://doi.org/10.1016/s0006-3495\(62\)86953-7](https://doi.org/10.1016/s0006-3495(62)86953-7) | PubMed | Google Scholar
2. Mel B (1991) **The Clusteron: Toward a Simple Abstraction for a Complex Neuron** 1991 Morgan-Kaufmann | Google Scholar
3. Poirazi P, Mel BW (2001) **Impact of active dendrites and structural plasticity on the memory capacity of neural tissue** *Neuron* **29**:779–96 [https://doi.org/10.1016/s0896-6273\(01\)00252-5](https://doi.org/10.1016/s0896-6273(01)00252-5) | PubMed | Google Scholar
4. Mel BW, Schiller J, Poirazi P (2017) **Synaptic plasticity in dendrites: complications and coping strategies** *Curr Opin Neurobiol* **43**:177–86 <https://doi.org/10.1016/j.conb.2017.03.012> | PubMed | Google Scholar
5. Kastellakis G, Poirazi P. (2019) **Synaptic Clustering and Memory Formation** *Front Mol Neurosci* **12** <https://doi.org/10.3389/fnmol.2019.00300> | PubMed | Google Scholar
6. Winnubst J, Lohmann C (2012) **Synaptic clustering during development and learning: the why, when, and how** *Front Mol Neurosci* **5** <https://doi.org/10.3389/fnmol.2012.00070> | PubMed | Google Scholar
7. Leighton AH, Lohmann C (2016) **The Wiring of Developing Sensory Circuits-From Patterned Spontaneous Activity to Synaptic Plasticity Mechanisms** *Front Neural Circuits* **10** <https://doi.org/10.3389/fncir.2016.00071> | PubMed | Google Scholar
8. Kirchner JH, Gjorgjieva J (2022) **Emergence of synaptic organization and computation in dendrites** *Neuroforum* **28**:21–30 <https://doi.org/10.1515/nf-2021-0031> | Google Scholar
9. Liang L, Chen C. (2020) **Organization, Function, and Development of the Mouse Retinogeniculate Synapse** *Annu Rev Vis Sci* **6**:261–85 <https://doi.org/10.1146/annurev-vision-121219-081753> | PubMed | Google Scholar
10. Bickford ME, Slusarczyk A, Dilger EK, Krahe TE, Kucuk C, Guido W (2010) **Synaptic development of the mouse dorsal lateral geniculate nucleus** *J Comp Neurol* **518**:622–35 <https://doi.org/10.1002/cne.22223> | PubMed | Google Scholar
11. Monavarfeshani A, Stanton G, Van Name J, Su K, Mills WA, Swilling K, Kerr A, Huebschman NA, Su J, Fox MA. (2018) **LRRTM1 underlies synaptic convergence in visual thalamus** *eLife* **7** <https://doi.org/10.7554/eLife.33498> | PubMed | Google Scholar
12. Hong YK, Park S, Litvina EY, Morales J, Sanes JR, Chen C (2014) **Refinement of the retinogeniculate synapse by bouton clustering** *Neuron* **84**:332–9 <https://doi.org/10.1016/j.neuron.2014.08.059> | PubMed | Google Scholar

13. Mason CA (1982) **Development of terminal arbors of retino-geniculate axons in the kitten-I. Light microscopical observations** *Neuroscience* **7**:541–59 [https://doi.org/10.1016/0306-4522\(82\)90063-x](https://doi.org/10.1016/0306-4522(82)90063-x) | PubMed | Google Scholar
14. Hamos JE, Van Horn SC, Raczkowski D, Sherman SM (1987) **Synaptic circuits involving an individual retinogeniculate axon in the cat** *J Comp Neurol* **259**:165–92 <https://doi.org/10.1002/cne.902590202> | PubMed | Google Scholar
15. Hammer S, Monavarfeshani A, Lemon T, Su J, Fox MA (2015) **Multiple Retinal Axons Converge onto Relay Cells in the Adult Mouse Thalamus** *Cell Rep* **12**:1575–83 <https://doi.org/10.1016/j.celrep.2015.08.003> | PubMed | Google Scholar
16. Morgan JL, Berger DR, Wetzel AW, Lichtman JW (2016) **The Fuzzy Logic of Network Connectivity in Mouse Visual Thalamus** *Cell* **165**:192–206 <https://doi.org/10.1016/j.cell.2016.02.033> | PubMed | Google Scholar
17. Budisantoso T, Matsui K, Kamasawa N, Fukazawa Y, Shigemoto R (2012) **Mechanisms Underlying Signal Filtering at a Multisynapse Contact** *J Neurosci* **32**:2357 <https://doi.org/10.1523/JNEUROSCI.5243-11.2012> | Google Scholar
18. Sherman SM, Guillery RW (1998) **On the actions that one nerve cell can have on another: distinguishing “drivers” from “modulators”** *Proc Natl Acad Sci U S A* **95**:7121–6 <https://doi.org/10.1073/pnas.95.12.7121> | PubMed | Google Scholar
19. Guillery RW, Sherman SM (2002) **Thalamic Relay Functions and Their Role in Corticocortical Communication: Generalizations from the Visual System** *Neuron* **33**:163–75 [https://doi.org/10.1016/S0896-6273\(01\)00582-7](https://doi.org/10.1016/S0896-6273(01)00582-7) | Google Scholar
20. Sretavan D, Shatz CJ (1984) **Prenatal development of individual retinogeniculate axons during the period of segregation** *Nature* **308**:845–8 <https://doi.org/10.1038/308845a0> | PubMed | Google Scholar
21. Sretavan DW, Shatz CJ (1986) **Prenatal development of retinal ganglion cell axons: segregation into eye-specific layers within the cat’s lateral geniculate nucleus** *J Neurosci* **6**:234–51 <https://doi.org/10.1523/JNEUROSCI.06-01-00234.1986> | PubMed | Google Scholar
22. Dhande OS, Hua EW, Guh E, Yeh J, Bhatt S, Zhang Y, Ruthazer ES, Feller MB, Crair MC (2011) **Development of single retinofugal axon arbors in normal and beta2 knock-out mice** *J Neurosci* **31**:3384–99 <https://doi.org/10.1523/JNEUROSCI.4899-10.2011> | PubMed | Google Scholar
23. Campbell G, Shatz CJ (1992) **Synapses formed by identified retinogeniculate axons during the segregation of eye input** *J Neurosci* **12**:1847–58 <https://doi.org/10.1523/JNEUROSCI.12-05-01847.1992> | PubMed | Google Scholar
24. Chen C, Regehr WG (2000) **Developmental remodeling of the retinogeniculate synapse** *Neuron* **28**:955–66 [https://doi.org/10.1016/S0896-6273\(00\)00166-5](https://doi.org/10.1016/S0896-6273(00)00166-5) | PubMed | Google Scholar
25. Hauser JL, Liu X, Litvina EY, Chen C (2014) **Prolonged synaptic currents increase relay neuron firing at the developing retinogeniculate synapse** *J Neurophysiol* **112**:1714–28 <https://doi.org/10.1152/jn.00451.2014> | PubMed | Google Scholar
26. Zhang C, Yadav S, Speer CM (2023) **The synaptic basis of activity-dependent eye-specific competition** *Cell Rep* **42**:112085 <https://doi.org/10.1016/j.celrep.2023.112085> | PubMed |

27. Vatan T, Minehart JA, Zhang C, Agarwal V, Yang J, Speer CM (2021) **Volumetric super-resolution imaging by serial ultrasectioning and stochastic optical reconstruction microscopy in mouse neural tissue** *STAR Protocols* **2**:100971 <https://doi.org/10.1016/j.xpro.2021.100971> | [Google Scholar](#)
28. Muir-Robinson G, Hwang BJ, Feller MB (2002) **Retinogeniculate axons undergo eye-specific segregation in the absence of eye-specific layers** *J Neurosci* **22**:5259–64 <https://doi.org/10.1523/jneurosci.22-13-05259.2002> | [PubMed](#) | [Google Scholar](#)
29. Xu HP, Burbridge TJ, Chen MG, Ge X, Zhang Y, Zhou ZJ, Crair MC (2015) **Spatial pattern of spontaneous retinal waves instructs retinotopic map refinement more than activity frequency** *Dev Neurobiol* **75**:621–40 <https://doi.org/10.1002/dneu.22288> | [PubMed](#) | [Google Scholar](#)
30. Xu HP, Furman M, Mineur YS, Chen H, King SL, Zenisek D, Zhou ZJ, Butts DA, Tian N, Picciotto MR, Crair MC (2011) **An instructive role for patterned spontaneous retinal activity in mouse visual map development** *Neuron* **70**:1115–27 <https://doi.org/10.1016/j.neuron.2011.04.028> | [PubMed](#) | [Google Scholar](#)
31. Xu HP, Burbridge TJ, Ye M, Chen M, Ge X, Zhou ZJ, Crair MC (2016) **Retinal Wave Patterns Are Governed by Mutual Excitation among Starburst Amacrine Cells and Drive the Refinement and Maintenance of Visual Circuits** *J Neurosci* **36**:3871–86 <https://doi.org/10.1523/JNEUROSCI.3549-15.2016> | [PubMed](#) | [Google Scholar](#)
32. Rossi FM, Pizzorusso T, Porciatti V, Marubio LM, Maffei L, Changeux JP (2001) **Requirement of the nicotinic acetylcholine receptor beta 2 subunit for the anatomical and functional development of the visual system** *Proc Natl Acad Sci U S A* **98**:6453–8 <https://doi.org/10.1073/pnas.101120998> | [PubMed](#) | [Google Scholar](#)
33. Grubb MS, Rossi FM, Changeux JP, Thompson ID (2003) **Abnormal functional organization in the dorsal lateral geniculate nucleus of mice lacking the beta 2 subunit of the nicotinic acetylcholine receptor** *Neuron* **40**:1161–72 [https://doi.org/10.1016/s0896-6273\(03\)00789-x](https://doi.org/10.1016/s0896-6273(03)00789-x) | [PubMed](#) | [Google Scholar](#)
34. Sun C, Warland DK, Ballesteros JM, van der List D, Chalupa LM (2008) **Retinal waves in mice lacking the beta2 subunit of the nicotinic acetylcholine receptor** *Proc Natl Acad Sci U S A* **105**:13638–43 <https://doi.org/10.1073/pnas.0807178105> | [PubMed](#) | [Google Scholar](#)
35. Stafford BK, Sher A, Litke AM, Feldheim DA (2009) **Spatial-temporal patterns of retinal waves underlying activity-dependent refinement of retinofugal projections** *Neuron* **64**:200–12 <https://doi.org/10.1016/j.neuron.2009.09.021> | [PubMed](#) | [Google Scholar](#)
36. Bansal A, Singer JH, Hwang BJ, Xu W, Beaudet A, Feller MB (2000) **Mice lacking specific nicotinic acetylcholine receptor subunits exhibit dramatically altered spontaneous activity patterns and reveal a limited role for retinal waves in forming ON and OFF circuits in the inner retina** *J Neurosci* **20**:7672–81 <https://doi.org/10.1523/jneurosci.20-20-07672.2000> | [PubMed](#) | [Google Scholar](#)
37. Hammer S, Carrillo GL, Govindaiah G, Monavarfeshani A, Bircher JS, Su J, Guido W, Fox MA (2014) **Nuclei-specific differences in nerve terminal distribution, morphology, and development in mouse visual thalamus** *Neural Dev* **9** <https://doi.org/10.1186/1749-8104-9-16> | [PubMed](#) | [Google Scholar](#)

38. Zhang Q, Lee WA, Paul DL, Ginty DD (2019) **Multiplexed peroxidase-based electron microscopy labeling enables simultaneous visualization of multiple cell types** *Nat Neurosci* **22**:828–39 <https://doi.org/10.1038/s41593-019-0358-7> | PubMed | Google Scholar
39. Koch SM, Dela Cruz CG, Hnasko TS, Edwards RH, Huberman AD, Ullian EM (2011) **Pathway-specific genetic attenuation of glutamate release alters select features of competition-based visual circuit refinement** *Neuron* **71**:235–42 <https://doi.org/10.1016/j.neuron.2011.05.045> | PubMed | Google Scholar
40. Su J, Byer L, Liang Y, Fox MA (2024) **Distribution, development, and identity of retinal ganglion cells labeled in the Sert-Cre reporter mouse** *Journal of Comparative Neurology* **532**:e25606 <https://doi.org/10.1002/cne.25606> | Google Scholar
41. Johnson KP, Fitzpatrick MJ, Zhao L, Wang B, McCracken S, Williams PR, Kerschensteiner D (2021) **Cell-type-specific binocular vision guides predation in mice** *Neuron* **109**:1527–39 <https://doi.org/10.1016/j.neuron.2021.03.010> | PubMed | Google Scholar
42. Assali A, Gaspar P, Rebsam A (2014) **Activity dependent mechanisms of visual map formation--from retinal waves to molecular regulators** *Semin Cell Dev Biol* **35**:136–46 <https://doi.org/10.1016/j.semcdb.2014.08.008> | PubMed | Google Scholar
43. Fassier C, Nicol X (2021) **Retinal Axon Interplay for Binocular Mapping** *Front Neural Circuits* **15** <https://doi.org/10.3389/fncir.2021.679440> | PubMed | Google Scholar
44. Penn AA, Riquelme PA, Feller MB, Shatz CJ (1998) **Competition in retinogeniculate patterning driven by spontaneous activity** *Science* **279**:2108–12 <https://doi.org/10.1126/science.279.5359.2108> | PubMed | Google Scholar
45. Assali A, Le Magueresse C, Bennis M, Nicol X, Gaspar P, Rebsam A (2017) **RIM1/2 in retinal ganglion cells are required for the refinement of ipsilateral axons and eye-specific segregation** *Sci Rep* **7**:3236 <https://doi.org/10.1038/s41598-017-03361-0> | PubMed | Google Scholar
46. Fredj NB, Hammond S, Otsuna H, Chien C-B, Burrone J, Meyer MP (2010) **Synaptic Activity and Activity-Dependent Competition Regulates Axon Arbor Maturation, Growth Arrest, and Territory in the Retinotectal Projection** *J Neurosci* **30**:10939 <https://doi.org/10.1523/JNEUROSCI.1556-10.2010> | Google Scholar
47. Hua JY, Smear MC, Baier H, Smith SJ (2005) **Regulation of axon growth in vivo by activity-based competition** *Nature* **434**:1022–6 <https://doi.org/10.1038/nature03409> | PubMed | Google Scholar
48. Rahman TN, Munz M, Kutsarova E, Bilash OM, Ruthazer ES (2020) **Stentian structural plasticity in the developing visual system** *Proc Natl Acad Sci U S A* **117**:10636–8 <https://doi.org/10.1073/pnas.2001107117> | PubMed | Google Scholar
49. Munz M, Gobert D, Schohl A, Poquérousse J, Podgorski K, Spratt P, Ruthazer ES (2014) **Rapid Hebbian axonal remodeling mediated by visual stimulation** *Science* **344**:904–9 <https://doi.org/10.1126/science.1251593> | PubMed | Google Scholar

50. Zhang LI, Tao HW, Holt CE, Harris WA (1998) **A critical window for cooperation and competition among developing retinotectal synapses** *Nature* **395**:37–44 <https://doi.org/10.1038/25665> | [Google Scholar](#)
51. Matsumoto N, Barson D, Liang L, Crair MC (2024) **Hebbian instruction of axonal connectivity by endogenous correlated spontaneous activity** *Science* **385**:eadh7814 <https://doi.org/10.1126/science.adh7814> | [PubMed](#) | [Google Scholar](#)
52. Yasuda M, Nagappan-Chettiar S, Johnson-Venkatesh EM, Umemori H (2021) **An activity-dependent determinant of synapse elimination in the mammalian brain** *Neuron* **109**:1333–49 <https://doi.org/10.1016/j.neuron.2021.03.006> | [PubMed](#) | [Google Scholar](#)
53. Louail A, Sierksma MC, Chaffiol A, Baudet S, Assali A, Couvet S, Nedjam M, Roche F, Zagar Y, Duebel J, Nicol X (2020) **cAMP-Dependent Co-stabilization of Axonal Arbors from Adjacent Developing Neurons** *Cell Rep* **33**:108220 <https://doi.org/10.1016/j.celrep.2020.108220> | [PubMed](#) | [Google Scholar](#)
54. Kutsarova E, Schohl A, Munz M, Wang A, Zhang YY, Bilash OM, Ruthazer ES (2023) **BDNF signaling in correlation-dependent structural plasticity in the developing visual system** *PLoS Biol* **21**:e3002070 <https://doi.org/10.1371/journal.pbio.3002070> | [PubMed](#) | [Google Scholar](#)
55. Godement P, Salaün J, Imbert M (1984) **Prenatal and postnatal development of retinogeniculate and retinocollicular projections in the mouse** *J Comp Neurol* **230**:552–75 <https://doi.org/10.1002/cne.902300406> | [PubMed](#) | [Google Scholar](#)
56. Mooney R, Madison DV, Shatz CJ (1993) **Enhancement of transmission at the developing retinogeniculate synapse** *Neuron* **10**:815–25 [https://doi.org/10.1016/0896-6273\(93\)90198-z](https://doi.org/10.1016/0896-6273(93)90198-z) | [PubMed](#) | [Google Scholar](#)
57. Ziburkus J, Dilger EK, Lo FS, Guido W (2009) **LTD and LTP at the developing retinogeniculate synapse** *J Neurophysiol* **102**:3082–90 <https://doi.org/10.1152/jn.90618.2008> | [PubMed](#) | [Google Scholar](#)
58. Butts DA, Kanold PO, Shatz CJ (2007) **A burst-based “Hebbian” learning rule at retinogeniculate synapses links retinal waves to activity-dependent refinement** *PLoS Biol* **5**:e61 <https://doi.org/10.1371/journal.pbio.0050061> | [PubMed](#) | [Google Scholar](#)
59. Lee H, Brott BK, Kirkby LA, Adelson JD, Cheng S, Feller MB, Datwani A, Shatz CJ (2014) **Synapse elimination and learning rules co-regulated by MHC class I H2-Db** *Nature* **509**:195–200 <https://doi.org/10.1038/nature13154> | [PubMed](#) | [Google Scholar](#)
60. Chung WS, Clarke LE, Wang GX, Stafford BK, Sher A, Chakraborty C, Joung J, Foo LC, Thompson A, Chen C, Smith SJ, Barres BA (2013) **Astrocytes mediate synapse elimination through MEGF10 and MERTK pathways** *Nature* **504**:394–400 <https://doi.org/10.1038/nature12776> | [PubMed](#) | [Google Scholar](#)
61. Stevens B, Allen NJ, Vazquez LE, Howell GR, Christopherson KS, Nouri N, Micheva KD, Mehalow AK, Huberman AD, Stafford B, Sher A, Litke AM, Lambris JD, Smith SJ, John SW, Barres BA (2007) **The classical complement cascade mediates CNS synapse elimination** *Cell* **131**:1164–78 <https://doi.org/10.1016/j.cell.2007.10.036> | [PubMed](#) | [Google Scholar](#)
62. Schafer DP, Lehrman EK, Kautzman AG, Koyama R, Mardinly AR, Yamasaki R, Ransohoff RM, Greenberg ME, Barres BA, Stevens B (2012) **Microglia sculpt postnatal neural circuits in an**

activity and complement-dependent manner *Neuron* **74**:691–705 <https://doi.org/10.1016/j.neuron.2012.03.026> | PubMed | Google Scholar

63. Hooks BM, Chen C (2006) **Distinct roles for spontaneous and visual activity in remodeling of the retinogeniculate synapse** *Neuron* **52**:281–91 <https://doi.org/10.1016/j.neuron.2006.07.007> | PubMed | Google Scholar
64. Dilger EK, Krahe TE, Morhardt DR, Seabrook TA, Shin HS, Guido W (2015) **Absence of plateau potentials in dLGN cells leads to a breakdown in retinogeniculate refinement** *J Neurosci* **35**:3652–62 <https://doi.org/10.1523/jneurosci.2343-14.2015> | PubMed | Google Scholar
65. Cork RJ, Namkung Y, Shin HS, Mize RR (2001) **Development of the visual pathway is disrupted in mice with a targeted disruption of the calcium channel beta(3)-subunit gene** *J Comp Neurol* **440**:177–91 <https://doi.org/10.1002/cne.1378> | PubMed | Google Scholar
66. Fernández-Nogales M, López-Cascales MT, Murcia-Belmonte V, Escalante A, Fernández-Albert J, Muñoz-Viana R, Barco A, Herrera E (2022) **Multiomic Analysis of Neurons with Divergent Projection Patterns Identifies Novel Regulators of Axon Pathfinding** *Adv Sci* **9**:e2200615 <https://doi.org/10.1002/advs.202200615> | PubMed | Google Scholar
67. Wang Q, Marcucci F, Cerullo I (2016) **Mason C. Ipsilateral and Contralateral Retinal Ganglion Cells Express Distinct Genes during Decussation at the Optic Chiasm** *eneuro* **3** <https://doi.org/10.1523/ENEURO.0169-16.2016> | Google Scholar
68. Nicol X, Voyatzis S, Muzerelle A, Narboux-Nême N, Südhof TC, Miles R, Gaspar P (2007) **cAMP oscillations and retinal activity are permissive for ephrin signaling during the establishment of the retinotopic map** *Nat Neurosci* **10**:340–7 <https://doi.org/10.1038/nn1842> | PubMed | Google Scholar
69. Balice-Gordon RJ, Chua CK, Nelson CC, Lichtman JW (1993) **Gradual loss of synaptic cartels precedes axon withdrawal at developing neuromuscular junctions** *Neuron* **11**:801–15 [https://doi.org/10.1016/0896-6273\(93\)90110-d](https://doi.org/10.1016/0896-6273(93)90110-d) | PubMed | Google Scholar
70. Gan WB, Lichtman JW (1998) **Synaptic segregation at the developing neuromuscular junction** *Science* **282**:1508–11 <https://doi.org/10.1126/science.282.5393.1508> | PubMed | Google Scholar
71. Wyatt RM, Balice-Gordon RJ (2003) **Activity-dependent elimination of neuromuscular synapses** *J Neurocytol* **32**:777–94 <https://doi.org/10.1023/b:Neur.0000020623.62043.33> | PubMed | Google Scholar
72. Chen J, Mizushige T, Nishimune H (2012) **Active zone density is conserved during synaptic growth but impaired in aged mice** *J Comp Neurol* **520**:434–52 <https://doi.org/10.1002/cne.22764> | PubMed | Google Scholar
73. Balice-Gordon RJ, Lichtman JW (1994) **Long-term synapse loss induced by focal blockade of postsynaptic receptors** *Nature* **372**:519–24 <https://doi.org/10.1038/372519a0> | PubMed | Google Scholar
74. Buffelli M, Burgess RW, Feng G, Lobe CG, Lichtman JW, Sanes JR (2003) **Genetic evidence that relative synaptic efficacy biases the outcome of synaptic competition** *Nature* **424**:430–4 <https://doi.org/10.1038/nature01844> | PubMed | Google Scholar

75. Kopp DM, Perkel DJ, Balice-Gordon RJ (2000) **Disparity in neurotransmitter release probability among competing inputs during neuromuscular synapse elimination** *J Neurosci* **20**:8771–9 <https://doi.org/10.1523/jneurosci.20-23-08771.2000> | PubMed | Google Scholar
76. Sanes JR, Lichtman JW (1999) **Development of the vertebrate neuromuscular junction** *Annu Rev Neurosci* **22**:389–442 <https://doi.org/10.1146/annurev.neuro.22.1.389> | PubMed | Google Scholar
77. Misgeld T, Kummer TT, Lichtman JW, Sanes JR (2005) **Agrin promotes synaptic differentiation by counteracting an inhibitory effect of neurotransmitter** *Proc Natl Acad Sci USA* **102**:11088–93 <https://doi.org/10.1073/pnas.0504806102> | Google Scholar
78. Lin W, Dominguez B, Yang J, Aryal P, Brandon EP, Gage FH, Lee KF (2005) **Neurotransmitter acetylcholine negatively regulates neuromuscular synapse formation by a Cdk5-dependent mechanism** *Neuron* **46**:569–79 <https://doi.org/10.1016/j.neuron.2005.04.002> | PubMed | Google Scholar
79. Hashimoto K, Kano M (2013) **Synapse elimination in the developing cerebellum** *Cell Mol Life Sci* **70**:4667–80 <https://doi.org/10.1007/s00018-013-1405-2> | PubMed | Google Scholar
80. Bosman LW, Konnerth A (2009) **Activity-dependent plasticity of developing climbing fiber-Purkinje cell synapses** *Neuroscience* **162**:612–23 <https://doi.org/10.1016/j.neuroscience.2009.01.032> | PubMed | Google Scholar
81. Hashimoto K, Kano M (2003) **Functional differentiation of multiple climbing fiber inputs during synapse elimination in the developing cerebellum** *Neuron* **38**:785–96 [https://doi.org/10.1016/s0896-6273\(03\)00298-8](https://doi.org/10.1016/s0896-6273(03)00298-8) | PubMed | Google Scholar
82. Nitta A, Yamasaki M, Miyazaki T, Konno K, Yoshimura H, Watanabe M (2025) **Molecular and Anatomical Strengthening of “Winner” Climbing Fiber Synapses in Developing Mouse Purkinje Cells** *J Neurosci* <https://doi.org/10.1523/JNEUROSCI.2156-24.2025> | PubMed | Google Scholar
83. Wilson AM, Schalek R, Suissa-Peleg A, Jones TR, Knowles-Barley S, Pfister H, Lichtman JW (2019) **Developmental Rewiring between Cerebellar Climbing Fibers and Purkinje Cells Begins with Positive Feedback Synapse Addition** *Cell Rep* **29**:2849–61 <https://doi.org/10.1016/j.celrep.2019.10.081> | Google Scholar
84. Bosman LWJ, Takechi H, Hartmann J, Eilers J, Konnerth A (2008) **Homosynaptic Long-Term Synaptic Potentiation of the “Winner” Climbing Fiber Synapse in Developing Purkinje Cells** *J Neurosci* **28**:798 <https://doi.org/10.1523/JNEUROSCI.4074-07.2008> | Google Scholar
85. Ohtsuki G, Hirano T (2008) **Bidirectional plasticity at developing climbing fiber-Purkinje neuron synapses** *Eur J Neurosci* **28**:2393–400 <https://doi.org/10.1111/j.1460-9568.2008.06539.x> | PubMed | Google Scholar
86. Lorenzetto E, Caselli L, Feng G, Yuan W, Nerbonne JM, Sanes JR, Buffelli M (2009) **Genetic perturbation of postsynaptic activity regulates synapse elimination in developing cerebellum** *Proc Natl Acad Sci USA* **106**:16475–80 <https://doi.org/10.1073/pnas.0907298106> | Google Scholar
87. Kawamura Y, Nakayama H, Hashimoto K, Sakimura K, Kitamura K, Kano M (2013) **Spike timing-dependent selective strengthening of single climbing fibre inputs to Purkinje cells during**

cerebellar development *Nat Commun* **4**:2732 <https://doi.org/10.1038/ncomms3732> | [Google Scholar](#)

88. Benninger K, Hood G, Simmel D, Tuite L, Wetzel A, Ropelewski A, Watkins S, Watson A, Bruchez M. (2020) **Cyberinfrastructure of a Multi-Petabyte Microscopy Resource for Neuroscience Research** In: *Practice and Experience in Advanced Research Computing* Portland, OR, USA: Association for Computing Machinery pp. 1–7 [Google Scholar](#)
89. Paxinos G. (2007) **Atlas of the developing mouse brain at E17.5** Amsterdam; Boston: Elsevier [Google Scholar](#)
90. Babcock H, Sigal YM, Zhuang X (2012) **A high-density 3D localization algorithm for stochastic optical reconstruction microscopy** *Optical Nanoscopy* **1** <https://doi.org/10.1186/2192-2853-1-6> | [Google Scholar](#)
91. Babcock HP, Huang F, Speer CM (2019) **Correcting Artifacts in Single Molecule Localization Microscopy Analysis Arising from Pixel Quantum Efficiency Differences in sCMOS Cameras** *Sci Rep* **9**:18058 <https://doi.org/10.1038/s41598-019-53698-x> | [PubMed](#) | [Google Scholar](#)

Author information

Chenghang Zhang

Department of Biology, University of Maryland, College Park, United States
ORCID iD: [0000-0003-1346-2186](https://orcid.org/0000-0003-1346-2186)

Tarlan Vatan

Department of Biology, University of Maryland, College Park, United States

Colenso M Speer

Department of Biology, University of Maryland, College Park, United States
ORCID iD: [0000-0002-3076-7072](https://orcid.org/0000-0002-3076-7072)

For correspondence: cspeer@umd.edu

Editors

Reviewing Editor

Marla Feller

University of California, Berkeley, Berkeley, United States of America

Senior Editor

Panayiota Poirazi

FORTH Institute of Molecular Biology and Biotechnology, Heraklion, Greece

Reviewer #1 (Public review):

Summary

The authors previously published a study of RGC boutons in the dLGN in developing wild-type mice and developing mutant mice with disrupted spontaneous activity. In the current manuscript, they have broken down their analysis of RGC boutons according to the number of Homer/Bassoon puncta associated with each vGlut3 cluster.

The authors find that, in the first post-natal week, RGC boutons with multiple active zones (mAZs) are about a third as common as boutons with a single active zone (sAZ). The size of the vGlut2 cluster associated with each bouton was proportional to the number of active zones present in each bouton. Within the author's ability to estimate these values ($n=3$ per group, 95% of results expected to be within ~ 2.5 standard deviations), these results are consistent across groups: 1) dominant eye vs. non-dominant eye, 2) wild-type mice vs. mice with activity blocked, and at 3) ages P2, P4, and P8. The authors also found that mAZs and sAZs also have roughly the same number (about 1.5) of sAZs clustered around them (within 1.5 μm).

However, the authors do not interpret this consistency between groups as evidence that active zone clustering is not a specific marker or driver of activity dependent synaptic segregation. Rather, the authors perform a large number of tests for statistical significance and cite the presence or absence of statistical significance as evidence that "Eye-specific active zone clustering underlies synaptic competition in the developing visual system (title)". I don't believe this conclusion is supported by the evidence.

Strengths

The source dataset is high resolution data showing the colocalization of multiple synaptic proteins across development. Added to this data is labeling that distinguishes axons from the right eye from axons from the left eye. The first order analysis of this data showing changes in synapse density and in the occurrence of multi-active zone synapses is useful information about the development of an important model for activity dependent synaptic remodeling.

Weaknesses

In my previous review I argued that it was not possible to determine, from their analysis, whether the differences they were reporting between groups was important to the biology of the system. The authors have made some changes to their statistics (paired t-tests) and use some less derived measures of clustering. However, they still fail to present a meaningfully quantitative argument that the observed group differences are important. The authors base most of their claims on small differences between groups. There are two big problems with this practice. First, the differences between groups appear too small to be biologically important. Second, the differences between groups that are used as evidence for how the biology works are generally smaller than the precision of the author's sampling. That is, the differences are as likely to be false positives as true positives.

(1) Effect size. The title claims: "Eye-specific active zone clustering underlies synaptic competition in the developing visual system". Such a claim might be supported if the authors found that mAZs are only found in dominant-eye RGCs and that eye-specific segregation doesn't begin until some threshold of mAZ frequency is reached. Instead, the behavior of mAZs is roughly the same across all conditions. For example, the clear trend in Figure 4C and D is that measures of clustering between mAZ and sAZ are as similar as could reasonably be expected by the experimental design. However, some of the comparisons of very similar values produced p -values < 0.05 . The authors use this fact to argue that the negligible differences between mAZ and sAZs explain the development of the dramatic differences in the distribution of ipsilateral and contralateral RGCs.

(2) Sample size. Performing a large number of significance tests and comparing p -values is not hypothesis testing and is not descriptive science. At best, with large sample sizes and

controls for multiple tests, this approach could be considered exploratory. With $n=3$ for each group, many comparisons of many derived measures, among many groups, and no control for multiple testing, this approach constitutes a random result generator.

The authors argue that $n=3$ is a large sample size for the type of high resolution / large volume data being used. It is true that many electron microscopy studies with $n=1$ are used to reveal the patterns of organization that are possible within an individual. However, such studies cannot control individual variation and are, therefore, not appropriate for identifying subtle differences between groups.

In response to previous critiques along these lines, the authors argue they have dealt with this issue by limiting their analysis to within-individual paired comparisons. There are several problems with their thinking in this approach. The main problem is that they did not change the logic of their arguments, only which direction they pointed the t-tests. Instead of claiming that two groups are different because $p < 0.05$, they say that two groups are different because one produced $p < 0.05$ and the other produced $p > 0.05$. These arguments are not statistically valid or biologically meaningful.

To the best of my understanding, the results are consistent with the following model:

- RGCs form mAZs at large boutons (known)
- About a quarter of week-one RGC boutons are mAZs (new observation)
- Vesicle clustering is proportional to active zone number (~new observation)
- RGC synapse density increases during the first post-week (known)
- Blocking activity reduces synapse density (known)
- Contralateral eye RGCs form more and larger synapses in the lateral dLGN (known)
- With $n=3$ and effect sizes smaller than 1 standard deviation, a statistically significant result is about as likely to be a false positive as a true positive.
- A true-positive statistically significant result does not provide evidence of a meaningful deviation from a biological model.

Providing plots that show the number of active zones present in boutons across these various conditions is useful. However, I could find no compelling deviation from the above default predictions that would influence how I see the role of mAZs in activity dependent eye-specific segregation.

Below are critiques of most of the claims of the manuscript.

Claim (abstract): individual retinogeniculate boutons begin forming multiple nearby presynaptic active zones during the first postnatal week.

Confirmed by data.

Claim (abstract): the dominant-eye forms more numerous mAZ contacts,

Misleading: The dominant-eye (by definition) forms more contacts than the non-dominant eye. That includes mAZ.

Claim (abstract): At the height of competition, the non-dominant-eye projection adds many single active zone (sAZ) synapses

Weak: While the individual observation is strong, it is a surprising deviation based on a single $n=3$ experiment in a study that performed twelve such experiments (six ages,

mutant/wildtype, sAZ/mAZ)

Claim (abstract): Together, these findings reveal eye-specific differences in release site addition during synaptic competition in circuits essential for visual perception and behavior.

False: This claim is unambiguously false. The above findings, even if true, do not argue for any functional significance to active zone clustering.

Claim (line 84): "At the peak of synaptic competition midway through the first postnatal week, the non-dominant-eye formed numerous sAZ inputs, equalizing the global synapse density between the two eyes"

Weak: At one of twelve measures (age, bouton type, genotype) performed with 3 mice each, one density measure was about twice as high as expected.

Claim (line 172): "In WT mice, both mAZ (Fig. 3A, left) and sAZ (Fig. 3B, left) inputs showed significant eye-specific volume differences at each age."

Questionable: There appears to be a trend, but the size and consistency is unclear.

Claim (line 175): "the median VGluT2 cluster volume in dominant-eye mAZ inputs was 3.72 fold larger than that of non-dominant-eye inputs (Fig. 3A, left)."

Cherry picking. Twelve differences were measured with an n of 3, 3 each time. The biggest difference of the group was cited. No analysis is provided for the range of uncertainty about this measure (2.5 standard deviations) as an individual sample or as one of twelve comparisons.

Claim (line 174): "In the middle of eye-specific competition at P4 in WT mice, the median VGluT2 cluster volume in dominant-eye mAZ inputs was 3.72 fold larger than that of non-dominant-eye inputs (Fig. 3A, left). In contrast, β 2KO mice showed a smaller 1.1 fold difference at the same age (Fig. 3A, right panel). For sAZ synapses at P4, the magnitudes of eye-specific differences in VGluT2 volume were smaller: 1.35-fold in WT (Fig. 3B, left) and 0.41-fold in β 2KO mice (Fig. 3B, right). Thus, both mAZ and sAZ input size favors the dominant eye, with larger eye-specific differences seen in WT mice (see Table S3)."

No way to judge the reliability of the analysis and trivial conclusion: To analyze effect size the authors choose the median value of three measures (whatever the middle value is). They then make four comparisons at the time point where they observed the biggest difference in favor of their hypothesis. There is no way to determine how much we should trust these numbers besides spending time with the mislabeled scatter plots. The authors then claim that this analysis provides evidence that there is a difference in vGluT2 cluster volume between dominant and non-dominant RGCs and that that difference is activity dependent. The conclusion that dominant axons have bigger boutons and that mutants that lack the property that would drive segregation would show less of a difference is very consistent with the literature. Moreover, there is no context provided about what 1.35 or 1.1 fold difference means for the biology of the system.

Claim (189): "This shows that vesicle docking at release sites favors the dominant-eye as we previously reported but is similar for like eye type inputs regardless of AZ number."

Contradicts core claim of manuscript: Consistent with previous literature, there is an activity dependent relative increase in vGluT2 clustering of dominant eye RGCs. The new information is that that activity dependence is more or less the same in sAZ and mAZ. The only plausible alternative is that vGluT2 scaling only increases in mAZ which would be consistent with the claims of their paper. That is not what they found. To the extent that the analysis presented in this manuscript tests a hypothesis, this is it. The claim of the title has been refuted by figure 3.

Claim (line 235): "For the non-dominant eye projection, however, clustered mAZ inputs outnumbered clustered sAZ inputs at P4 (Fig. 4C, bottom left panel), the age when this eye adds sAZ synapses (Fig. 2C)."

Misleading: The overwhelming trend across 24 comparisons is that the sAZ clustering looks like mAZ clustering. That is the objective and unambiguous result. Among these 24 underpowered tests (n=3), there were a few p-values < 0.05. The authors base their interpretation of cell behavior on these crossings.

Claim (line 328): "The failure to add synapses reduced synaptic clustering and more inputs formed in isolation in the mutants compared to controls."

Trivially true: Density was lower in mutant.

Claim (line 332): "While our findings support a role for spontaneous retinal activity in presynaptic release site addition and clustering..."

Not meaningfully supported by evidence: I could not find meaningful differences between WT and mutant beside the already known dramatic difference in synapse density.

<https://doi.org/10.7554/eLife.91431.3.sa3>

Reviewer #2 (Public review):

Summary:

In this manuscript, Zhang and Speer examine changes in the spatial organization of synaptic proteins during eye specific segregation, a developmental period when axons from the two eyes initially mingle and gradually segregate into eye-specific regions of the dorsal lateral geniculate. The authors use STORM microscopy and immunostain presynaptic (VGluT2, Bassoon) and postsynaptic (Homer) proteins to identify synaptic release sites. Activity-dependent changes of this spatial organization are identified by comparing the β 2KO mice to WT mice. They describe two types of synapses based on Bassoon clustering: the multiple active zone (mAZ) synapse and single active zone (sAZ) synapse. In this revision, the authors have added EM data to support the idea that mAZ synapses represent boutons with multiple release sites. They have also reanalyzed their data set with different statistical approaches.

Strengths:

The data presented is of good quality and provides an unprecedented view at high resolution of the presynaptic components of the retinogeniculate synapse during active developmental remodeling. This approach offers an advance to the previous mouse EM studies of this synapse because of the CTB label allows identification of the eye from which the presynaptic terminal arises.

Weaknesses:

While the interpretation of this data set is much more grounded in this second revised submission, some of the authors' conclusions/statements still lack convincing supporting evidence. In particular, the data does not support the title: "Eye-specific active zone clustering underlies synaptic competition in the developing visual system". The data show that there are fewer synapses made for both contra- and ipsi- inputs in the β 2KO mice-- this fact alone can account for the differences in clustering. There is no evidence linking clustering to synaptic competition. Moreover, the findings of differences in AZ# or distance between AZs that the authors report are quite small and it is not clear whether they are functionally meaningful.

Reviewer #3 (Public review):

This study is a follow-up to a recent study of synaptic development based on a powerful data set that combines anterograde labeling, immunofluorescence labeling of synaptic proteins, and STORM imaging (Cell Reports, 2023). Specifically, they use anti-Vglut2 label to determine the size of the presynaptic structure (which they describe as the vesicle pool size), anti-Bassoon to label active zones with the resolution to count them, and anti-Homer to identify postsynaptic densities. Their previous study compared the detailed synaptic structure across the development of synapses made with contra-projecting vs. ipsi-projecting RGCs and compared this developmental profile with a mouse model with reduced retinal waves. In this study, they produce a new detailed analysis on the same data set in which they classify synapses into "multi-active zone" vs. "single-active zone" synapses and assess the number and spacing of these synapses. The authors use measurements to make conclusions about the role of retinal waves in the generation of same-eye synaptic clusters. The authors interpret these results as providing insight into how neural activity drives synapse maturation, the strength of their conclusions is not directly tested by their analysis.

Strengths:

This is a fantastic data set for describing the structural details of synapse development in a part of the brain undergoing activity-dependent synaptic rearrangements. The fact that they can differentiate the eye of origin is what makes this data set unique over previous structural work. The addition of example images from the EM dataset provides confidence in their categorization scheme.

Weaknesses:

Though the descriptions of single vs multi-active zone synapses are important and represent a significant advance, the authors continue to make unsupported conclusions regarding the biological processes driving these changes. Although this revision includes additional information about the populations tested and the tests conducted, the authors do not address the issue raised by previous reviews. Specifically, they provide no assessment of what effect size represents a biologically meaningful result. For example, a more appropriate title is "The distribution of eye-specific single vs multi-active zone is altered in mice with reduced spontaneous activity" rather than concluding that this difference in clustering is somehow related to synaptic competition. Of course, the authors are free to speculate, but many of the conclusions of the paper are not supported by their results.

<https://doi.org/10.7554/eLife.91431.3.sa1>

Author response:

The following is the authors' response to the previous reviews

Reviewer #1 (Public review)

Summary:

This manuscript addresses the question of whether spontaneous activity contributes to the clustering of retinogeniculate synapses before eye opening. The authors re-analyze a previously published dataset to answer the question. The authors conclude that synaptic clustering is eye-specific and activity dependent during the first postnatal week. While

there is useful information in this manuscript, I don't see how the data meaningfully supports the claims made about clustering.

In adult retinogeniculate connections, functional specificity is supported by select pairings of retinal ganglion cells and thalamocortical cells forming dozens of synaptic connections in subcellular microcircuits called glomeruli. In this manuscript, the authors measure whether the frequency of nearby synapses is higher in the observed data than in a model where synapses are randomly distributed throughout the volume. Any real anatomical data will deviate from such a model. The interesting biological question is not whether a developmental state deviates from random. The interesting question is how much of the adult clustering occurs before eye opening. In trying to decode the analysis in this manuscript, I can't tell if the answer is 99% or 0.001%.

We thank the reviewer for their helpful critique through both rounds of review. We have refocused the manuscript on paired eye-specific measurements of active zone addition and spatial relationships among active zones at each age. All effect sizes and power values for each comparison are now reported in Table S2. These measures allow readers to gauge biological significance more transparently.

Strengths:

The source dataset is high resolution data showing the colocalization of multiple synaptic proteins across development. Added to this data is labeling that distinguishes axons from the right eye from axons from the left eye. The first order analysis of this data showing changes in synapse density and in the occurrence of multi-active zone synapses is useful information about the development of an important model system.

Weaknesses:

I don't think the analysis of clustering within this dataset improves our understanding of how the system works. It is possible that the result is clear to the authors based on looking at the images. As a reader trying to interpret the analysis, I ran into the following problems:

- *It is not possible to estimate biologically meaningful effect sizes from the data provided. Spontaneous activity in the post natal week could be responsible for 99% or 0.001% of RGC synapse clustering.*
- *The sample size is too small for the kinds of comparisons being made. The authors point out that many STORM studies use an n of 1 while the authors have $n = 3$ for each of their six experimental groups. However, the critical bit is what kinds of questions you are trying to answer with a given sample size. This study depends on determining whether the differences between groups are due to age, genotype, or individual variation. This study also makes multiple comparisons of many different noisy parameters that test the same or similar hypothesis. In this context, it is unlikely that $n = 3$ sufficiently controls for individual variation.*

We have revised the manuscript to focus on eye-specific differences, which are paired measurements collected at each age. We have measured effect sizes and performed power tests for all comparisons presented in the manuscript. These measurements are shown for every figure in a new supplemental table S2.

- *There is no clear biological interpretation of the core measure of the publication, the normalized clustering index. The normalized clustering index starts with counting the fraction of single active zone synapses within various distances to the edge of synapses. This frequency is compared to a randomization model in which the positions of synapses*

are randomized throughout a volume. The authors found that the biggest deviation between the observed and randomized proximity frequency using a distance threshold of 1.5 μ m. They consider the deviation from the random model to be a sign of clustering. However, two RGC synapses 1.5 μ m apart have a good chance of coming from the same RGC axon. At this scale, real observations will, therefore, always look more clustered than a model where synapses are randomly placed in a volume. If you randomly place synapses on an axon, they will be much closer together than if you randomly place synapses within a volume. The authors normalize their clustering measure by dividing by the frequency of clustering in the normalized model. That makes the measure of clustering an ambiguous mix of synapse clustering, axon morphology, and synaptic density.

We have removed the “normalized clustering index”. “Clustered” inputs are now defined strictly as those that have a neighboring single active-zone (sAZ) synapse within 1.5 μ m. For each type of input (sAZ and mAZ) we show 1) the ratio of clustered to isolated inputs for both eyes, and 2) the number of neighboring sAZs (Figure 4).

We agree with the reviewer that many synapses are likely made nearby along the same axon from an individual RGC. In this scenario, sAZ synapses that are nearby a neighboring mAZ input may be part of the same nascent bouton. And, sAZ synapses nearby other sAZ neighbors may ultimately mature into a mAZ input. At the same time, inputs from one RGC may form nearby other inputs from neighboring RGCs. We discuss these motifs and potential mechanisms of cell-autonomous and non-autonomous development (Lines 300-308).

• Other measures are also very derived. For instance, one argument is based on determining that the cumulative distribution of the distance of dominant-eye multi-active zone synapses with nearby single-active zone synapses from dominant-eye multi-active zone synapses is statistically different from the cumulative distribution of the distance of dominant-eye multi-active zones without nearby single-active zone synapses from dominant-eye multi-active zones. Multiple permutations of this measure are compared.

We have simplified the presentation to show all measured path lengths for every input. This allows the reader to see each of the inputs and their relative distances. We present these data for like-eye type interactions at P4 and P8 (Figures 5 and S5).

• There are major biological differences between groups that are difficult to control for. Between P2, P4, and P8, there are changes in cell morphology and synaptic density. There are also large differences in synapse density between wild type and KO mice. It is difficult to be confident that these differences are not responsible for the relatively subtle changes in clustering indices.

• Many claims are based on complicated comparisons between groups rather than the predominating effects within the data. It is noted that: "In KO mice, dominant eye projections showed increased clustering around mAZ synapses compared to sAC synapses suggesting partial maintenance of synaptic clustering despite retinal wave defects". In contrast, I did not notice any discussion of the fact that the most striking trend in those measures is that the clustering index decreases from P2 to P8.

Related to the points above, we have revised the manuscript to focus on eye-specific release site addition and spatial relationships. For clarity, we have removed the clustering index and instead present ratios of clustered and isolated inputs, the number of sAZ synapses near each input type, and distance between like-eye mAZ inputs (Figure 4).

• Statistics are improperly applied. In my first review I tried to push the authors to calculate confidence intervals for two reasons. First, I believed the reader should be able

to answer questions such as whether 99% or 0.01% of RGC synaptic clustering occurred in the first postnatal week. Second, I wanted the authors to deal with the fact that $n=3$ is underpowered for many of the questions they were asking. While many confidence intervals can now be found leading up to a claim, it is difficult to find claims that are directly supported by the correct confidence interval. Many claims are still incorrectly based on which combinations of comparisons produced statistically significant differences and which combinations did not.

We have substantially revised the manuscript to focus on within-group paired effects between eye-of-origin. We performed power tests for all statistical presentations and effect sizes and powers are presented for every figure in a new supplemental table S2. To simplify the manuscript and make it easier to read, we report confidence interval measurements in a separate supplemental table S3.

Reviewer #2 (Public review):

Summary:

This study provides a valuable data set showing changes in the spatial organization of synaptic proteins at the retinogeniculate connection during a developmental period of active axonal and synaptic remodeling. The data collected by STORM microscopy is state-of-the-art in terms of the high-resolution view of the presynaptic components of a plastic synapse. The revision has addressed many, but not all, of the initial concerns about the authors interpretation of their data. However, with the revisions, the manuscript has become very dense and difficult to follow.

We greatly appreciate the reviewer's thoughtful comments through two rounds of review. To improve the clarity of the manuscript, we have substantially revised the work to streamline the narrative, clearly define terminology, and simplify data presentations, allowing readers to more directly interpret results and their implications.

Strengths:

The data presented is of good quality and provides an unprecedented view at high resolution of the presynaptic components of the retinogeniculate synapse during active developmental remodeling. This approach offers an advance to the previous mouse EM studies of this synapse because the CTB label allows identification of the eye from which the presynaptic terminal arises.

Weaknesses:

From these data the authors conclude that eye-specific increase in mAZ synapse density occur over retinogeniculate refinement, that sAZ synapses cluster close to mAZ synapses over age, and that this process depends on spontaneous activity and proximity to eye-specific mAZ synapses. While the interpretation of this data set is much more grounded in this revised submission, some of the authors' conclusions/statements still lack convincing supporting evidence.

This includes:

(1) The conclusion that multi-active zone synapses are loci for synaptic clustering. This statement, or similar ones (e.g., line 407) suggest that mAZ synapses actively or through some indirect way influence the clustering of sAZ synapses. There is no evidence for this. Clustering of retinal synapses are in part due to the fact that retinal inputs synapse on the proximal dendrites. With increased synaptogenesis, there will be increased density of

retinal terminals that are closely localized. And with development, perhaps sAZ synapses mature into mAZ synapses. This scenario could also explain a large part of this data set.

We thank the reviewer for their comment. We have removed the ambiguous phrasing and clarified the manuscript to explicitly discuss alternative interpretations consistent with the results (Lines 300-308). This includes a discussion of sAZ synapse maturation into mAZ inputs (Lines 294-296).

(2) The conclusion that, "clustering depends on spontaneous retinal activity" could be misleading to the reader given that the authors acknowledge that their data is most consistent with a failure of synaptogenesis in the mutant mice (in the rebuttal). Additionally clustering does occur in CTB+ projections around mAZ synapses.

We have removed the highlighted phrase and revised the manuscript to focus on differences in release site addition between eye-of-origin. We clarified our discussion of activity-dependent changes to state that synapses fail to form in the mutant and synaptic clustering was reduced (Lines 324-330).

(3) Line 403: "Since mAZ synapses are expected to have a higher release probability, they likely play an important role in driving plasticity mechanisms reliant on neurotransmission." What evidence do the authors have that mAZ are expected to have higher release probability?

We thank the reviewer for their careful reading. Because they have several active zones, mAZ synapses are expected to have a higher number of release sites (N), which could be independent of release probability at any individual active zone (Pr). We have removed the reference to release probability. Instead, we maintain focus on active zone number.

Reviewer #3 (Public review):

This study is a follow-up to a recent study of synaptic development based on a powerful data set that combines anterograde labeling, immunofluorescence labeling of synaptic proteins, and STORM imaging (Cell Reports, 2023). Specifically, they use anti-Vglut2 label to determine the size of the presynaptic structure (which they describe as the vesicle pool size), anti-Bassoon to label active zones with the resolution to count them, and anti-Homer to identify postsynaptic densities. Their previous study compared the detailed synaptic structure across the development of synapses made with contra-projecting vs. ipsi-projecting RGCs and compared this developmental profile with a mouse model with reduced retinal waves. In this study, they produce a new detailed analysis on the same data set in which they classify synapses into "multi-active zone" vs. "single-active zone" synapses and assess the number and spacing of these synapses. The authors use measurements to make conclusions about the role of retinal waves in the generation of same-eye synaptic clusters, providing key insight into how neural activity drives synapse maturation.

Strengths:

This is a fantastic data set for describing the structural details of synapse development in a part of the brain undergoing activity-dependent synaptic rearrangements. The fact that they can differentiate eye of origin is what makes this data set unique over previous structural work. The addition of example images from EM data set provides confidence in their categorization scheme.

Weaknesses:

Though the descriptions of synaptic clusters are important and represent a significant advance, the authors conclusions regarding the biological processes driving these clusters are not testable by such a small sample. This limitation is expected given the massive effort that goes into generating this data set. Of course the authors are free to speculate, but many of the conclusions of the paper are not statistically supported.

We thank the reviewer for their helpful comments throughout the revision process. We have substantially modified the manuscript to reframe the work around release site addition during eye-specific competition. Power tests and effect size measurements are presented for every figure in a new supplemental table S2.

Reviewer #2 (Recommendations for the authors):

(1) Authors should discuss that it is not clear what the relationship is between sAZ and mAZ, and sAZ could turn into a mAZ. This is not unreasonable that the number of AZ/bouton increases with development given that in the adult rodent retinogeniculate bouton, there is an average of 27 active zones (Budisantoso et al, 2012).

We thank the reviewer for their helpful suggestion. We have added a discussion of the relationship between sAZ and mAZ inputs and the point that sAZ synapses may mature into mAZ synapses (Lines 294-296). We now reference the work of Budisantoso et al., J. Neurosci. 2012.

(2) The authors should clarify how the statistics are calculated for the normalized clustering index (figure 3B, C). For ratios of values each with variance, the variance is summed when calculating SEM.

For clarity, we have removed the normalized clustering index analysis. We have simplified the work to present a clear definition of clustered and unclustered inputs, where clustering is defined by the presence of a nearby neighboring synapse within 1.5mm. We present the ratio of clustered and unclustered inputs for each input type and eye-of-origin. We also show the number of sAZ synapses nearby each clustered input (Figure 4).

(3) The authors have significantly clarified the terminology that they use in the text. This is much appreciated. However, it would be helpful to the naïve reader if they could define their use of the word "synapse" as referring to individual active zones/release sites or to terminals/boutons. For example:

Line 378: "Prior electron microscopy studies in the mouse found limited evidence of convergent synaptic clustering from neighboring RGCs at postnatal day 8 (10, 13), suggesting that the mAZ synapses seen in STORM images are single retinogeniculate terminals. The lack of synaptic convergence in prior EM reconstructions at P8 implies that early clustering around mAZ synapses may result from local output clustering within individual RGC arbors."

What do the authors mean by "convergent synaptic clustering": do they mean clustering of release sites from different RGC inputs? And what does "local output clustering" mean?

We thank the reviewer for their suggestion to use clear terminology. We have revised the manuscript to define our use of the term "synapse" as a single active zone/release site (Lines 134-136). We refer to mAZ boutons in STORM data as "inputs". We have revised the discussion of prior EM studies (Lines 130-132) and clarified all discussions of synaptic clustering throughout the work.

(4) While the authors argue that the retina-specific $\beta 2$ -nAChR mice exhibit disrupted retinal waves and defects in eye specific segregation, the authors are studying issues of active zone density which may depend on mechanisms depending on the postsynaptic neuron. This should be acknowledged.

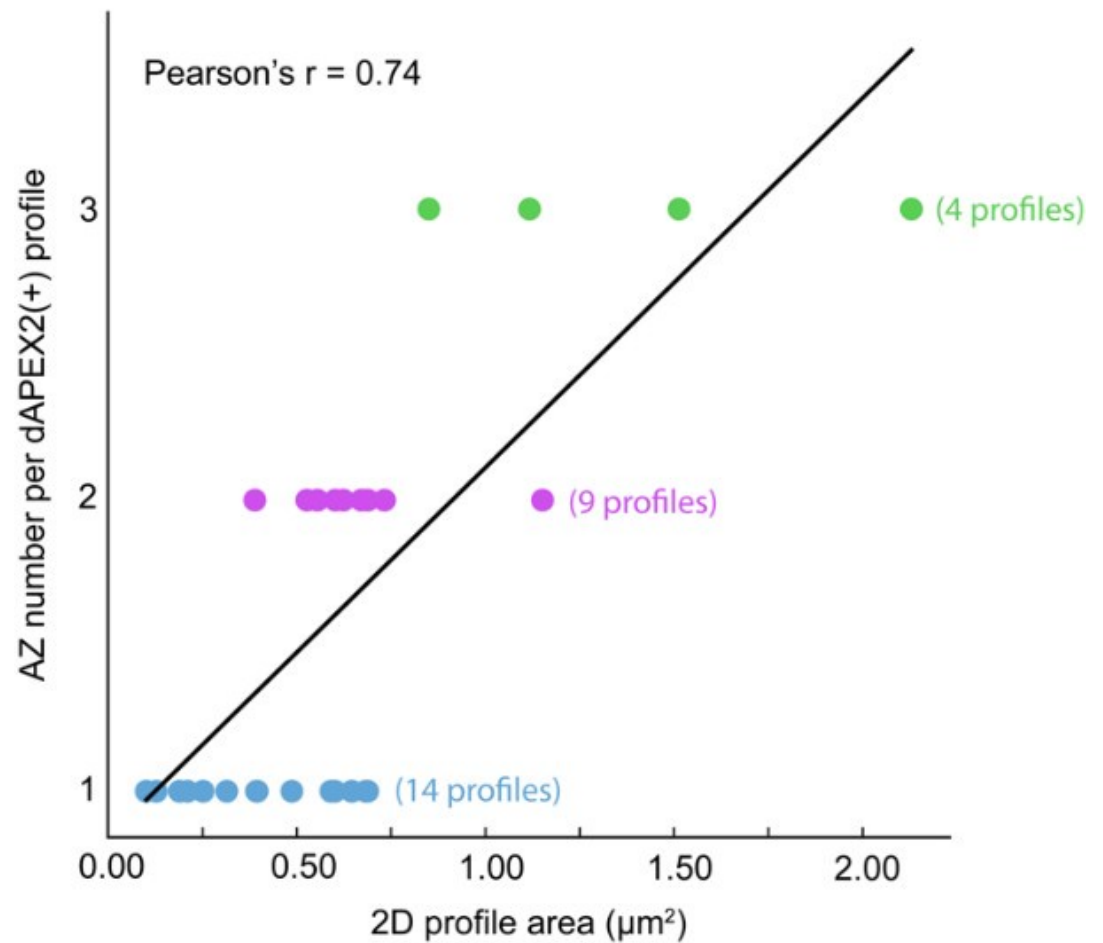
We have updated the text to discuss the fact that postsynaptic mechanisms are also critical for the refinement of eye-specific synapses (Lines 332-340). We have added several additional references to the manuscript accordingly.

Reviewer #3 (Recommendations for the authors):

The authors have addressed many of my original concerns. The additional description of criteria for categorizing synapses, showing all the data points, gives the reader a stronger sense of where the numbers in the quantification come from. Replacing the "complex/simple" distinction with the "multi/single active zone" and the other clarifying text was effective. The addition of the EM data was also a very nice example to help interpret STORM images. It does appear there was no quantification on this EM data set and perhaps just a few example images were taken as "proof of principle". If, by chance, the authors have more EM images to make a data set of them that allows for some quantification, that would be great to add.

We thank the reviewer for their helpful comments on the manuscript through both rounds of review. The EM data we collected were 2D images of a subset of physical sections at postnatal day 8. Most dAPEX2(+) profiles had a single active zone, but a definitive identification would require 3D imaging so that each terminal can be assessed in its entirety for release sites that might be missed in a single cross section. Similarly, multi-active zone boutons are positively identified in 2D images, but definitive measurements of AZ number would require 3D information. We analyzed our 2D EM images and present a plot of dAPEX2(+) profile size versus active zone number below. These measures are positively correlated ($r = 0.74$), with larger profiles containing more active zones.

Author response image 1.



Unfortunately, we are not currently equipped to perform volumetric EM imaging at our home institution and are concerned that analysis of 2D data may be inconclusive. For these reasons, we are opting to maintain a qualitative presentation of our current EM results and we look forward to collaborating with other experts to achieve volumetric EM reconstructions in the future

<https://doi.org/10.7554/eLife.91431.3.sa0>